WHY MACHIAVELLI WORKS

The Organizational Admissibility of Political Power

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PART I — WOLVES, SHEEP, AND SHEPHERDS

CHAPTER 1 — THE EPISODE

On summer grazing allotments in Oregon, a sheep producer runs a band of approximately 1,000 ewes and lambs after the snow has melted and the public lands become accessible. The allotments include land administered by the U.S. Forest Service and the Bureau of Land Management. A full-time sheepherder manages the band with livestock guardian dogs. According to a 2024 technical report on carnivore-livestock conflict management, predation events on this operation had historically occurred at night.

Before wolves arrived on the landscape in 2014, the producer already gathered the sheep near camp overnight. At that time, the stated purpose was to keep the sheep close to camp. Once wolves became present, the purpose changed. Nighttime had become the period of greatest vulnerability, and preventing nighttime depredation became an explicit management objective.

The producer adopted electrified night penning. The pen had to constrain movement enough to keep sheep from separating excessively and becoming easier targets, but it could not simply be made as small as possible. It had to leave enough room for sheep to lie comfortably and avoid conditions in which lambs could be laid on. Heavy rain or mud could require relocation to reduce disease risk.

The fencing was not used alone. Livestock guardian dogs complemented the night pen. Often one dog stayed inside with the sheep while others remained outside, maintaining a perimeter presence. Human proximity was also part of the arrangement.

The reported result was striking but not final. The producer reported zero nighttime sheep depredations after incorporating electrified night penning. After four to five years, however, the producer observed an increase in daytime depredation incidents. The technical report states that predators changed their activity patterns in response to the management practice. The changed timing also changed the producer's response: wolves were easier to haze during daylight.

The intervention had not eliminated wolves, predation risk, or the need for management. It had altered the conditions under which wolves could reach sheep and the circumstances under which people and dogs could respond.

CHAPTER 2 — WHERE IS MACHIAVELLI?

At first the answer appears almost embarrassingly obvious. Machiavelli seems to occupy the position of the shepherd.

The shepherd does not ask whether the wolf is wicked. He asks what the wolf does. He does not begin with what ought to protect the sheep. He observes what actually protects

them. He identifies a vulnerable condition, changes that condition, retains defensive capacity, and responds again when circumstances change.

At this stage, these parallels are recognitions rather than derivations. Chapters VI, XII–XV, XX, and XXV provide possible textual points of contact: the preservation of new orders after initial adherence weakens; dependence upon internally retained defensive capacity; attention to the verità effettuale rather than imagined political orders; the condition-relative usefulness of defensive measures; and preparation against changes attributed to fortuna. But the Oregon episode has not established that these passages preserve a common organizational relation. That relation, if present, must be derived from Machiavelli's text rather than projected backward from the flock.

Read provisionally, the Oregon episode appears almost naturally Machiavellian: do not govern according to how the world ought to behave; govern according to how it actually behaves.

CHAPTER 3 — THE FAMILIAR AMORAL READING

The familiar interpretation can now be made in its strongest form. Preservation may require measures that appear objectionable when judged apart from the conditions that make them effective.

Suppose the sheep would prefer not to enter the pen. The fence restricts their locally available movement. The dogs may disturb them. The shepherd determines where they remain during the period of greatest danger. Yet the immediate answer is formidable: the sheep are alive.

Within the flock example, the governing intervention and the continuation of the governed organization appear closely aligned. An individual sheep need not produce the collective organization required to reduce nighttime vulnerability. The governing capacity imposes a relation that the sheep do not independently produce, and that relation contributes to continuation of the flock.

From here a familiar Machiavellian inference becomes tempting: preservation requires constraint; therefore the governing power must be permitted the constraints preservation requires.

But even before that inference is evaluated, its structure should be noticed. It moves from a constraint that contributed to a specified continuation under specified conditions to a general permission claimed by the governing power. It also assumes, without yet examining the assumption, that preservation of the governing capacity and preservation of the governed organization remain aligned.

This is where Machiavelli's reputation for amorality becomes intelligible. The question seems to have shifted from whether a measure is conventionally good to whether it works. Necessity appears capable of overruling ordinary objection. What has not yet been

determined is whose continuation supplies that necessity, under what conditions it holds, and whether effectiveness, necessity, and admissibility have been silently identified.

CHAPTER 4 — THE ADMISSIBILITY CALCULUS (AC) INTERVENTION

But something has happened too quickly. Before asking whether the constraint is moral, Admissibility Calculus asks a more elementary question: what exactly has been demonstrated?

Let X_o denote the observed protective arrangement, including electrified night penning, guardian dogs, and human proximity. Let C_F denote the specified continuation of the flock, and K_w the active-predation conditions under which nighttime separation creates vulnerability. Let E(X, C, K) mean that X is effective for continuation C under conditions K, and let N(X, C, K) mean that X is necessary for continuation C under conditions K.

The reported observation supports an effectiveness claim:

$$\mathbf{E(X_o, C_F, Kw)}$$

The observed protective arrangement was associated with the elimination of reported nighttime depredations under the specified conditions.

It does not by itself establish:

$$\mathbf{N(X_o, C_F, K_w)}$$

Necessity would require showing that removal of a contribution supplied by X_o, while the relevant continuation and conditions are retained, eliminates a dependency required for C_F. The report does not provide that demonstration. Nor does it isolate the contribution of the electrified boundary from those of the dogs, human proximity, nighttime gathering, and their interaction.

Three questions must therefore be kept separate. Effectiveness asks whether X contributes to C under K. Necessity asks whether a dependency supplied by X is required for C under K. Admissibility asks whether retaining X, in its actual form and degree, is warranted relative to C, the stated conditions, other dependencies, available alternatives, and the relevant horizon.

Effectiveness does not entail necessity.

Necessity at one degree does not entail necessity at every greater degree.

The Oregon episode itself warns against collapsing these questions. The pen had to constrain movement, but it could not simply be made smaller without limit. Rain and mud could change the consequences of confinement. The wolves changed their activity. The conditions under which the intervention was evaluated did not remain fixed.

CHAPTER 5 — THE MISSING CONTINUATION

The story has so far encouraged an unnoticed identification. The continuation of the shepherd's governing capacity and the continuation of the flock appear to be the same thing. In the factual Oregon episode, that approximation is reasonable: the management measures are being used to keep sheep from being killed.

But organizationally the continuations remain distinguishable.

C_F = flock continuation

C_G = continued availability and exercise of the governing capacity

Necessity for C_G does not establish necessity for C_F.

A measure can be required for C_G without being required for C_F. The fact that governing capacity was established or retained because it contributes to flock continuation does not make every condition favorable to the governing capacity a condition necessary to the flock.

The distinction becomes visible if the analytical model is varied after the factual episode has been established. Suppose a threat disappears but a restriction remains because it makes the flock easier to govern. Suppose a governing mechanism suppresses a capacity of the flock because that capacity makes governance more difficult. Suppose preservation of governing authority begins to require conditions that degrade the organization it was established to preserve.

$N(X, C_G, K)$ does not entail $N(X, C_F, K)$

A measure does not become necessary for C_F merely because it contributes to C_G.

The moral question has not disappeared. It has been preceded by a specification question: for which continuation, under which conditions, has the claimed necessity been established?

CHAPTER 6 — RETURN TO MACHIAVELLI

Now return to *The Prince*.

When Machiavelli says that a prince must do something, what continuation is being preserved? The answer cannot be supplied silently by the reader. It must be derived from the text.

Sometimes continuation of the prince, continuation of the political order, and continuation of the governed organization may contribute strongly to one another. Sometimes they may coincide for the problem under analysis. But they need not be identical.

Let C_p designate continuation of the prince's possession or control of the state, and C_o continuation of a containing political organization within which that control operates.

N(X, C_P, K) does not entail N(X, C_O, K).

This does not refute Machiavelli. It specifies him.

Machiavelli may analyze with considerable precision what preserves a particular political continuation. A reader may then evaluate the resulting measure relative to a different continuation: continued availability of constitutional government, institutional legitimacy, civil liberty, population welfare, long-term state capacity, or another containing political organization. The reader may consequently judge the measure inadmissible even where Machiavelli has correctly identified its effectiveness relative to C_p.

The discrepancy may be real. But before judging it, the continuations must be distinguished. Otherwise an analysis of what preserves C_P is silently treated as though it had claimed to preserve every continuation contained within or affected by C_P.

This is the first point at which the familiar moral reputation of Machiavelli can be reconsidered without rehabilitating him. The question is no longer simply whether Machiavelli is moral or amoral. It is whether some of his apparent necessity claims are indexed to a political continuation that readers—and sometimes Machiavelli himself—may leave insufficiently specified.

CHAPTER 7 — THE METHODOLOGICAL TURN

The wolves, sheep, shepherd, dogs, and fence have done their expository work. They have not established anything about Machiavelli. They have shown why specification matters and why an apparently obvious claim of protective necessity can conceal multiple continuations.

The load-bearing investigation must now begin from Machiavelli's text itself.

Machiavelli's text → textually supported relation → indexed continuation and conditions → organizational reconstruction → AC Weakening → stopping condition → frozen proposition → heterogeneous cross-domain evaluation → result

The analogy may reveal a candidate relation. It may make that relation intelligible. It may show why the relation is worth testing. It cannot establish that Machiavelli supplied it.

The candidate relation must first be demonstrably supported by *The Prince*. Only then may political particulars be progressively removed. Only after the resulting proposition has reached a stated stopping condition and been frozen may it be evaluated against another political case, a heterogeneous intentional organization, or a case in which intentional agency is absent. Only after the resulting relation is frozen may it be tested

against other political cases, heterogeneous intentional organizations, and eventually cases in which intentional agency disappears entirely.

The shepherd therefore provides the entrance to the investigation, not its conclusion.

CHAPTER 8 — ANALYSIS

Original organizational reconstruction

Under conditions of active predation, continuation of a flock may depend upon a governing capacity able to alter the relations through which wolves gain access to individual sheep. The wolves need not destroy the flock to succeed. Continuation of predation requires only sufficient access to individual prey; it does not require destruction of the flock.

Individual sheep may respond through locally available actions such as flight, bunching, or immediate avoidance. Such actions may be intelligible relative to the immediate situation of the individual sheep without preserving either that individual or the flock containing it.

The shepherd can alter the conditions under which predation succeeds. Fences can reduce accessibility, night pens can reduce exposure and separation, dogs can increase the cost of penetration and provide detection, and human presence can provide additional detection and intervention. These measures can constrain actions locally available to individual sheep while nevertheless contributing to continuation of the flock.

The effectiveness of the observed protective arrangement does not by itself establish its necessity. Let X_o denote the composite arrangement of electrified night penning, guardian dogs, human proximity, and nighttime gathering. The reported outcome supports $E(X_o, C_F, K_W)$, but it does not establish $N(X_o, C_F, K_W)$. Establishing necessity would require showing that removing a contribution supplied by X_o while retaining the relevant continuation and conditions eliminates a dependency required for C_F. Nor may the result established for X_o be attributed separately to the fence, dogs, human proximity, or any other constituent without separate evaluation.

Necessity therefore remains indexed to conditions and continuation. A boundary necessary under active nighttime predation does not remain necessary merely because it was necessary previously. Nor does increasing a protective constraint necessarily increase protection. A stronger boundary may interfere with other dependencies required by the same continuation.

The governing capacity also possesses a continuation distinguishable from continuation of the flock. Let C_G designate continuation of the governing capacity. A measure necessary for C_G is not thereby necessary for C_F. Even where governing capacity contributes to flock continuation, $N(X, C_G, K)$ does not entail $N(X, C_F, K)$.

The organizational problem therefore cannot be reduced to whether the shepherd protects the sheep. It requires distinguishing the threatening continuation, the threatened continuation, and the continuation of the governing capacity, together with the conditions and dependencies relative to which necessity is claimed.

An AC Weakening

Remove wolves, sheep, shepherds, fences, dogs, pens, predation, prey acquisition, nighttime exposure, and the pastoral setting.

A threatened organization may confront another continuation whose pursuit exploits or produces conditions that degrade dependencies required by the threatened organization. Destruction of the threatened organization need not be the objective of the threatening continuation; sufficient exploitation of accessible dependencies may be enough for its own continuation.

Parts of the threatened organization may possess locally available actions that are intelligible relative to their immediate conditions without preserving either those parts or the organization containing them.

A governing capacity may alter the dependencies through which disruption succeeds. It may do so by changing accessibility, exposure, detection, constraint, defensive capacity, response, or other relations relevant to continuation. Such intervention may require constraining actions locally available to organizational parts.

The effectiveness of an intervention does not establish its necessity, and necessity under one set of conditions does not establish necessity under another. A necessity claim N(X, C, K) remains indexed to the continuation C and conditions K for which removal of X produces the relevant consequence. Increasing X does not necessarily increase preservation of C, because the intervention may alter other dependencies required by the same continuation.

The continuation of a governing capacity is distinguishable from the continuation for which that capacity is exercised. A measure necessary for continuation of the governing capacity is therefore not established as necessary or admissible for the governed organization merely because governing capacity contributes to its preservation.

The retained structure for this analysis therefore includes the distinction among a threatening continuation, a threatened continuation, and the continued availability of a governing capacity, together with the conditions, dependencies, locally available actions, and interventions through which those continuations affect one another. Further weakening would have to preserve enough of these relations to explain why the intervention can contribute to one continuation while constraining another.

Weaker still

A continuation may depend upon constraining locally available actions without those constraints being admissible merely because they preserve the constraining organization.

What is necessary must therefore be indexed not only to the threat being resisted, but to the continuation whose preservation makes the resistance necessary.

CHAPTER 9 — SOURCE ATTRIBUTION AND TEXTUAL ANCHORS

Matthew Collins et al., Reducing Risk on the Range: Non-Lethal Practices for Managing Carnivore-Livestock Conflicts (Western Landowners Alliance, November 2024), especially the Oregon electrified-night-penning case study and accompanying night-penning guidance. The factual account in Sections 1–4 is derived principally from this case study.

Suzanne A. Stone et al., "Adaptive Use of Nonlethal Strategies for Minimizing Wolf-Sheep Conflict in Idaho," Journal of Mammalogy 98, no. 1 (2017): 33–44. DOI: 10.1093/jmammal/gyw188. This study provides separate evidence concerning adaptive nonlethal wolf-management strategies; it does not document the Oregon operation described above.

Niccolò Machiavelli, The Prince, especially Chapters VI, XII–XV, XX, and XXV. At this stage these chapters supply provisional textual points of contact rather than an established common derivation. The load-bearing derivations begin in Part III and remain answerable to Machiavelli's text and the project's documented Lucid conversion.

PART II — THE METHOD

CHAPTER 1 — WHY MACHIAVELLI STILL WORKS

Machiavelli's political particulars are extraordinarily remote from many modern circumstances: hereditary principalities, mercenary armies, papal territorial power, fortified cities, and dynastic acquisition. Yet readers continue to find some of his observations analytically applicable beyond those circumstances.

Conventional explanations are available. Perhaps relevant features of human conduct have changed less than political institutions. Perhaps power politics continues to reproduce similar strategic problems. Perhaps Machiavelli was unusually perceptive about rulers, dependencies, and political behavior.

This project investigates a different but compatible possibility:

Some of Machiavelli's observations continue to be analytically useful because the organizational relations expressed through their political particulars are less particular than the particulars themselves.

In this project, saying that Machiavelli "works" does not mean that his prescriptions are correct, universally applicable, morally admissible, or uniquely insightful. It means that a relation supportable from his political analysis may remain coherent and explanatorily useful after some of the historical and political particulars through which he expressed it have been removed.

This project does not propose that organizational portability exhausts Machiavelli's continuing force. His rhetoric, political psychology, historical position, moral provocation, and treatment of violence may remain independently explanatory. The investigation isolates one narrower question: whether relations reconstructed from his political analyses retain analytical content after those specifically political and historical features have been progressively removed.

At this stage, that remains a hypothesis to be investigated rather than a conclusion.

The crucial distinction is between recognizing that a modern situation resembles Renaissance politics and determining whether a relation reconstructed from Machiavelli remains evaluable after its Renaissance political particulars have been progressively removed.

The first is analogy.

The second is the problem investigated here through textual reconstruction, AC Weakening, and heterogeneous cross-domain evaluation.

CHAPTER 2 — WHAT THIS PROJECT CLAIMS—AND DOES NOT CLAIM

This project reconstructs selected relations from *The Prince*, progressively weakens them by removing particulars, identifies a stopping condition, freezes the resulting propositions, and evaluates them in structurally heterogeneous organizations.

The evaluations are independent of Machiavelli's historical domain in the limited sense that the garden, cancer treatment, CCIBASIC, measurement standards, technological formats, artistic practices, evolution, and ecology do not belong to Renaissance political organization. They are not independently selected in the stronger experimental sense of random sampling, blinded selection, or selection without prior awareness of their possible relevance. Cross-domain survival in these cases therefore demonstrates applicability under heterogeneous organizations; it does not constitute independent empirical confirmation.

The project does not claim that Machiavelli consciously formulated the resulting generalized propositions.

It does not claim that Machiavelli anticipated cancer treatment, software architecture, measurement standards, technological format adoption, natural selection, or ecology.

It does not claim that successful cross-domain instantiation establishes a universal law or demonstrated organizational invariant.

It does not claim that the surviving relations are unique to Machiavelli or that another political thinker could not yield comparable relations under the same procedure.

It does not claim that cross-domain survival by itself explains Machiavelli's historical influence or continuing reputation. It investigates whether organizational survival can account for part of his continuing analytical usefulness.

It does not claim that AC validates Machiavelli's political prescriptions or establishes their moral admissibility.

The notation used in the investigation records distinctions, indices, and proposed dependencies. It does not by itself prove that a dependency exists. The warrant for a reconstructed relation must come from Machiavelli's text, and the warrant for its cross-domain applicability must come from the organization being evaluated.

The term calculus names the broader AC[1] project. This volume uses its specification, distinction, and weakening operations; it does not present AC as a complete deductive proof system.

[1] Admissibility Calculus, Zenodo Concept DOIs: 10.5281/zenodo.21982239 (AC I-IV) and 10.5281/zenodo.22240655 (AC IV)

The direction of reconstruction matters. The garden is not evidence from which Machiavelli's analysis of Louis XII is reconstructed. Machiavelli's text supplies the initial political relation; AC Weakening removes selected particulars while preserving what the reconstruction treats as its relevant organizational structure; and the resulting proposition is frozen before it is evaluated in the garden. Any further relation first exposed by the garden belongs initially to that evaluation and may not be attributed backward to Machiavelli without separate textual support.

Non-instantiation must also remain an admissible result. If a case lacks a relation required by the frozen proposition, that absence must be reported rather than concealed by altering the proposition or redescribing the case merely to produce a fit.

The Hyperrealism/Expressionism evaluation illustrates a boundary of this kind. Difference between the two pictorial organizations does not establish competition between them. Competition appears only after additional conditions make their continuations mutually consequential. The initial non-instantiation is not reversed by the later specification of those conditions; the additional specification identifies the relation absent from the original case.

This does not make Hyperrealism/Expressionism an independent falsification of the method. It is a boundary analysis showing that a cross-domain organization need not instantiate a frozen relation merely because the relevant vocabulary can be applied to it.

CHAPTER 3 — THE SPECIFICATION PROBLEM

The wolves, sheep, and shepherds example exposes the basic specification problem before the full method is introduced.

A statement such as "X is necessary" is organizationally underspecified if the continuation for which X is claimed to be necessary and the conditions under which the claim applies remain unstated.

Now let N(X, C, K) mean that X is necessary for continuation C under conditions K.

Part I distinguishes continuation of the flock, C_F, from continued availability and exercise of the governing capacity, C_G. It therefore establishes that:

N(X, C_G, K) does not entail N(X, C_F, K).

The same problem appears when Machiavelli is introduced. Let C_p designate continuation of the prince's possession or control of the state, and C_o continuation of a containing political organization within which that control operates. Then:

N(X, C_p, K) does not entail N(X, C_o, K).

The problem extends beyond necessity.

Let E(X, C, K) mean that X is effective for continuation C under conditions K, and let A(X, C, K) mean that X is admissible for continuation C under conditions K. Then:

E(X, C, K) does not entail N(X, C, K),

and:

N(X, C, K) does not entail A(X, C, K).

Effectiveness identifies a contribution. Necessity identifies a dependency whose removal prevents the specified continuation under the retained conditions. Admissibility asks whether the intervention is warranted relative to the specified continuation, conditions, alternatives, affected dependencies, and relevant horizon.

For consistency with AC IV, let $S = \langle C, B, K, H \rangle$ designate an Organizational Specification consisting of continuation C, organizational boundary B, conditions K, and horizon H. The abbreviated forms E(X, C, K), N(X, C, K), and A(X, C, K) used in this investigation suppress B and H only where they are held fixed or are not material to the immediate distinction. Their omission does not license those elements to vary silently. A change in C, B, K, or H produces a distinguishable specification, and a result established under one specification may not be transferred to another without further derivation.

Nor can necessity be transferred silently between continuations:

$N(X, C_1, K)$ does not entail $N(X, C_2, K)$.

Conditions must also remain indexed:

$N(X, C, K_1)$ does not entail $N(X, C, K_2)$.

The same specification discipline may apply to other apparently free-standing descriptions, including *successful*, *independent*, *competitive*, *preserving*, and *harmful*. Their organizational significance may change with the continuation, conditions, organizational level, or horizon relative to which they are evaluated.

Containing organizations create an especially important form of the specification problem. A component or competing organization may possess a local continuation C_L while participating within a containing organization possessing continuation C_o. An intervention or strategy may contribute to C_L while reconstructing K in ways that degrade dependencies required for C_o.

There is no contradiction in saying that an action is locally effective and systemically destructive when the two evaluations are indexed to different continuations.

The specification question therefore precedes any unqualified evaluative conclusion:

For what continuation, under what conditions, is the claimed relation being asserted?

Two qualifications are important.

First, "X is necessary" is not literally incomplete in ordinary speech; its continuation and conditions may be implicitly understood. It is *organizationally underspecified* for AC evaluation.

Second, it avoids saying that an action "degrades C_o." An intervention affects conditions or dependencies and thereby reduces the continuability of C_o. The continuation is the result being preserved, not ordinarily the object directly damaged.

CHAPTER 4 — DERIVATION AND AC WEAKENING

The method used in Part III can be represented as a provenance sequence:

source text → textually supported relation → indexed continuation and conditions → organizational reconstruction → AC Weakening → stopping condition → frozen proposition → cross-domain evaluation → result

The arrows in this sequence indicate procedural order and provenance. They do not represent formal logical implication. The notation used throughout the investigation records distinctions and proposed dependencies; the evidential warrant for those dependencies must come from the source text or the evaluated case.

Each stage constrains what may be claimed at the next.

Textual derivation. The candidate relation must first be supportable from Machiavelli's text. A relation discovered in a modern or cross-domain case cannot be projected backward into *The Prince*. Where The Prince LUCID's organizational notes assist in identifying the relation, that contribution should remain distinguishable from the Machiavellian passage that supplies the textual evidence.

Indexing. The continuation and conditions under which the candidate relation holds must be stated before weakening. If the source supports effectiveness relative to C_1 under K_1, the analysis may not silently reconstruct that result as necessity, admissibility, or effectiveness relative to C_2 or K_2.

Organizational reconstruction. The historically particular case is articulated in terms of the actors, continuations, conditions, dependencies, interventions, and consequences required to preserve the relation being investigated. Reconstruction makes the candidate structure explicit; it does not yet establish that every reconstructed element can be removed.

AC Weakening in this investigation is an application-level reconstruction procedure. It is informed by, but is not identical to, Admissible Weakening within the foundational AC derivation. Foundational Admissible Weakening removes organizational commitment while preserving a Retained Organizational Objective and continued derivability. Here the initial object is instead a relation derived from an external source text. Particular commitments are removed only while the source-warranted relation and its specification remain preserved. The resulting proposition is a disciplined interpretive reconstruction eligible for cross-domain evaluation, not a theorem derived within AC itself.

AC Weakening. Particular commitments are progressively removed while asking whether the remaining proposition still preserves the relation supported by the preceding derivation. Princes, armies, territory, political objectives, intentional action, and other particulars may disappear only if their removal does not eliminate or alter a dependency required by that relation.

Weakening is subtractive with respect to organizational commitment, but it may require rearticulation. Replacing "prince" with "governing organization," for example, changes the vocabulary in which the relation is expressed while attempting to remove the specifically political commitment. Such rearticulation is admissible only if it does not add a relation unavailable in the preceding derivation.

Stopping condition. Weakening must stop when further removal would eliminate a dependency required to explain the source result or leave a proposition too weak to discriminate among cases. The objective is not the weakest sentence that can possibly remain true. It is a minimally articulated proposition, relative to the documented weakening path, that preserves the relevant relation and identifies conditions whose absence would produce non-instantiation.

The stopping condition does not establish that the proposition is absolutely the weakest formulation available under every possible weakening path. It establishes only that the next proposed removal would no longer preserve the selected relation.

Freeze. Once a proposition has been selected for cross-domain evaluation, its wording and retained relations are frozen. The cross-domain case may not determine what the proposition was allowed to say.

Freezing is an internal control against altering the proposition after encountering the case. It is not equivalent to preregistration, blinded case selection, or independent experimental design. Because the cases in this investigation were deliberately selected for their

analytical value, freezing limits post hoc alteration of the proposition but does not make the selection of the cases independent.

Cross-domain evaluation. A structurally heterogeneous organization is introduced and evaluated against the frozen proposition. The evaluation asks whether the relations retained in the proposition are instantiated in the new organization, which required relations are absent, and whether the case exposes a consequence not contained in the frozen proposition.

The possible results must be distinguished.

Instantiation: the evaluated organization contains the relations required by the frozen proposition.

Non-instantiation: one or more required relations are absent from the evaluated organization.

Index failure: an apparent instantiation disappears when the relevant continuation, conditions, organizational level, or horizon is specified correctly.

Boundary identification: the evaluation identifies an additional condition under which a previously absent relation becomes instantiated.

Counterinstance: the evaluated organization satisfies the stated conditions of a proposition but fails to exhibit a consequence that the proposition requires. This result is available only where the frozen proposition makes a sufficiently strong conditional claim.

Non-instantiation is not the same as falsification. If a case lacks a relation required by the frozen proposition, the result shows that the proposition does not instantiate there; it does not by itself show that the proposition is false. Conversely, merely finding a case in which the proposition can be instantiated does not establish that the proposition is universal, invariant, or uniquely Machiavellian.

The direction of the procedure can be represented compactly as:

$$\mathbf{M \rightarrow D \rightarrow W \rightarrow F \rightarrow T \rightarrow R}$$

where M is Machiavelli's source text, D the derived relation, W the weakened relation, F the frozen proposition, T the cross-domain evaluation, and R the result exposed by that evaluation. Again, the arrows represent provenance and procedural sequence rather than logical entailment.

Two methodological restrictions then become essential.

The no-rescue rule. A frozen proposition may not be broadened, narrowed, or redescribed merely to force successful instantiation in a cross-domain case.

If a case T_0 lacks a relation required by F, the result remains:

T_0 does not instantiate F.

Suppose an additional condition Q is then introduced and the newly specified case T_1 instantiates F. The result is:

$T_1 = T_0$ under Q instantiates F.

This does not convert the original non-instantiation into success. It identifies Q as a condition under which the relation becomes available.

The Hyperrealism/Expressionism analysis has this form. Difference between the two pictorial organizations does not by itself instantiate competition. When a containing organization allocating scarce resources is introduced, a competitive relation can arise. The additional condition identifies the relation absent from the original specification; it does not erase the original non-instantiation or constitute a falsification and rescue of the frozen proposition.

The no-backward-attribution rule. A consequence first exposed by a cross-domain evaluation may delimit the domain of the frozen proposition or motivate a successor proposition. It may not be inserted retroactively into the proposition that was tested, and it may not be attributed backward to Machiavelli without separate textual support.

If F evaluated under T produces result R, then R belongs initially to that evaluation. A successor proposition F′ may later be constructed from F and R, but F′ must be identified as a downstream development. The result originally obtained under F remains unchanged.

If the garden reveals that unspecified continuations can nevertheless be affected, that result belongs initially to the garden evaluation. If CCIBASIC reveals that capability can be redistributed without equivalent redistribution of expertise, that result belongs initially to the CCIBASIC evaluation. If ecology reveals that relational organization need not imply common objective, that result belongs initially to the ecological evaluation.

Those results may justify returning to Machiavelli and asking whether corresponding relations can be established separately in his text. They cannot be credited to him merely because they emerged downstream from propositions originally derived from his political analysis.

The complete inferential discipline is therefore asymmetric:

Machiavelli's text supplies D.

D is weakened into and frozen as F.

F is evaluated under T, producing R.

But:

T may not retroactively alter F merely to produce instantiation.

And:

R may not be attributed to Machiavelli without separate textual support.

This asymmetry permits the cross-domain evaluations to perform genuine analytical work without converting their downstream discoveries into claims that Machiavelli said, intended, anticipated, or established more than his text supports.

PART III — THE TESTS

CHAPTER 1 — SAVONAROLA AND ALEXANDER VI

Chapter VI: the unarmed prophet. Machiavelli uses Fra Girolamo Savonarola as the contemporary counterexample to founders capable of preserving new orders after persuasion ceases to be sufficient. In the Lucid conversion:

"In our own time, this is what happened to Friar Girolamo Savonarola. He was ruined together with his new institutions as soon as the multitude began to cease believing in him. He possessed no means of holding firm those who had believed, or of making the unbelievers believe."

The historical record sharpens but also complicates Machiavelli's compression.

Savonarola's influence in Florence depended substantially upon prophetic credibility, popular adherence, organized supporters, and his position within the Florentine political and religious environment after the expulsion of the Medici. His campaigns for moral reform, including the collection and burning of objects condemned as "vanities," became highly visible expressions of his movement. They also supplied his opponents with conspicuous targets around which resistance could be organized. They should not, however, be treated as a sufficient explanation of his fall.

Pope Alexander VI supplied a different source of pressure. Savonarola's attacks upon ecclesiastical corruption and resistance to papal commands led to restrictions upon his preaching, excommunication in 1497, and the threat that Florence could face an interdict if it continued to protect him. Papal pressure therefore increased the political and economic cost to Florence of maintaining its alignment with Savonarola. Alexander did not simply remove Savonarola through external force; he altered the conditions under which Savonarola's internal support had to continue.

The crisis culminated in the proposed ordeal by fire of April 1498. Fra Domenico da Pescia accepted a Franciscan challenge on Savonarola's behalf. After prolonged disputes and delay, the ordeal did not occur. The assembled public received no confirming sign, and the frustrated crowd directed substantial blame toward Savonarola and his supporters. His convent was attacked, he was arrested, and on May 23, 1498, he and two Dominican companions were executed.

These events acted upon different relations supporting Savonarola's continuation. His reform practices could polarize adherence. Papal pressure increased the cost of political protection. Florentine opponents could exploit both. The failed ordeal then damaged the prophetic credibility that Machiavelli explicitly identifies as a principal source of Savonarola's authority.

The result should nevertheless be stated carefully. Savonarola's supporters and some institutions associated with the Florentine republic survived him. His fall therefore did not eliminate every organization he had influenced. What collapsed was more specifically Savonarola's personal authority and the continued governing availability of the reform movement organized around his prophetic leadership.

WMW is consequently not treating Machiavelli's compressed account as a complete historical explanation. It is asking whether the dependency Machiavelli identifies—initial acquisition through belief without an adequately retained means of preservation—can be reconstructed from the case without accepting every detail of his historical judgment.

The Prince Cross-References

Chapter VI — New principalities acquired by one's own arms and *virtù*. This is the controlling passage. Machiavelli distinguishes obtaining belief from preserving a new order after belief becomes unstable. His diagnosis is not merely that Savonarola ceased to persuade. It is that Savonarola lacked an adequately retained means of preserving his continuation when persuasion ceased to be sufficient.

Chapter VII — Cesare Borgia and foundations. Machiavelli praises Borgia's attempt to convert power initially received through his father and *fortuna* into foundations retained under his own control. Savonarola presents an inverse dependency: the relation that initially made his authority available was not converted into another relation capable of preserving that authority after prophetic adherence weakened.

Chapter XI — Ecclesiastical principalities and Alexander VI. This chapter provides a contextual contrast rather than the primary derivation. Machiavelli presents Alexander as using money, arms, and political opportunity to enlarge the Church's temporal power. Savonarola therefore confronted an organization increasingly able to impose material and political costs while his own authority remained substantially dependent upon continued belief and Florentine protection.

Chapter XIII — One's own arms and the arms of others. The controlling problem is retained availability. Capability supplied through another organization can be effective without becoming securely available to the continuation that depends upon it. Savonarola's dependence upon belief is not identical to dependence upon foreign arms, but both cases raise the question whether a continuation has retained the relations required to preserve itself when the conditions of its initial success change.

An AC Weakening

Savonarola's failure need not be converted into a general rule that political innovators require coercive force.

Let C_s designate the continued governing availability of Savonarola's reform movement under his prophetic leadership. Let K_1 designate conditions under which sufficient prophetic belief, popular adherence, and political protection remain available. Let K_2 designate the altered conditions under which those relations have weakened.

Under K_1, prophetic credibility and voluntary adherence contributed substantially to C_s. Preaching, civic mobilization, moral reform, and political influence operated within a surrounding relation in which enough Florentines continued to treat Savonarola's authority as credible and supportable.

The transition from K_1 to K_2 did not remove every believer or every institution Savonarola had influenced. It reduced the availability of the relations through which his own authority and movement had been coordinated. Papal pressure made political protection more costly. Organized opponents exploited declining support. The failed ordeal further damaged prophetic credibility.

The relevant question is therefore not whether Savonarola possessed arms in a literal sense. It is whether some adequately retained relation remained available to preserve C_s after the relation that had initially produced adherence ceased to be sufficient.

AC Weakening: An organization initially established through one relation need not preserve itself through that same relation. But if the initial relation ceases to be sufficient, continuation requires some other adequately retained relation through which the organization can remain available.

Machiavelli's "arms" identify one historically powerful preservation mechanism. They do not establish that coercion is the only possible mechanism. Savonarola's structural vulnerability, as Machiavelli presents it, is that no adequate substitute preserved his authority when belief and political protection ceased to do so.

Weaker Still

Remove Florence, prophecy, political authority, and coercive force.

Weaker still: If an organization's continuation depends upon a relation whose continued availability the organization cannot preserve, then the organization remains vulnerable to the loss of that relation, regardless of how effective the relation was in producing the organization initially.

This is the retained structure beneath Machiavelli's contrast between armed and unarmed prophets.

Acquisition and preservation need not have the same dependencies. A relation can be sufficient to establish an organization without being sufficient to preserve it under changed conditions. The relevant question is therefore not only whether a relation previously worked, but whether the conditions that made it effective remain available—

or whether another adequate relation has been retained for continuation after those conditions change.

Savonarola provides a particularly clear instance of Machiavelli's diagnosis. Belief contributed to the establishment of his authority. Internal opposition and papal pressure altered the conditions under which that authority operated. The failed ordeal further damaged prophetic credibility. Machiavelli identifies no adequately retained preservation relation capable of compensating for that loss.

This does not independently prove that Machiavelli's account exhausts the historical causes of Savonarola's fall. It shows that his account can be reconstructed as a dependency analysis rather than merely as praise of force.

Analytical Significance

Chapter VI participates in a recurrent Machiavellian distinction between an advantage that is presently available and a foundation through which that advantage remains available under changed conditions.

Cesare Borgia attempts to convert power received through his father and *fortuna* into foundations of his own. Auxiliary arms supply effective capacity while preserving dependence upon the organization that controls them. Alexander VI converts spiritual position, money, arms, and political opportunity into increased temporal capacity. Savonarola, by contrast, remains dependent upon a relation of belief and adherence that he cannot adequately preserve or replace when conditions change.

These are not identical cases, and WMW need not claim that they instantiate one complete principle. They share a narrower question:

Does the organization retain the relation required for its continuation when the conditions of its initial success change?

The Savonarola episode therefore provides more than an illustration of Machiavelli's maxim that unarmed prophets fail. It exposes a distinction between initial acquisition and retained preservation capacity.

That distinction is the chapter's source-derived result. It is not yet a cross-domain confirmation.

Source Note

Primary textual source: Niccolò Machiavelli, *The Prince*, Chapter VI. Chapters VII and XIII provide supporting internal parallels concerning externally dependent capacity and retained foundations. Chapter XI provides the principal Machiavellian context for

Alexander VI's temporal power. The quoted passage follows the project's Lucid conversion.

Historical sources:

- Donald Weinstein, Savonarola: The Rise and Fall of a Renaissance Prophet (New Haven: Yale University Press, 2011).
- Lauro Martines, *Fire in the City: Savonarola and the Struggle for the Soul of Renaissance Florence* (New York: Oxford University Press, 2006).
- John T. Scott, "The Fortune of Machiavelli's Unarmed Prophet," *The Journal of Politics* 80, no. 2 (2018): 615–629, DOI: 10.1086/696992. This is especially relevant because it examines Savonarola's wider role within *The Prince*, rather than treating the Chapter VI reference as an isolated maxim.

CHAPTER 2 — POLITICAL EXTENSION: U.S. PRESIDENTIAL ELECTIONS

Original Description

Consider a modern U.S. presidential general election through a Machiavellian lens. Although other candidates may participate, two leading campaign organizations ordinarily dominate the competition. They engage through advertising, rallies, debates, interviews, targeted appeals, opposition research, social-media communication, surrogate advocacy, and strategically timed disclosures.

The immediate objective of each campaign is candidate-specific electoral victory. Let C_a designate victory by candidate A and C_b victory by candidate B. Under an election in which only one candidate can assume the office:

C_a and C_b are mutually exclusive local continuations.

Each campaign attempts to increase favorable evaluation of its own candidate and to reduce favorable evaluation of its opponent. Information is therefore a principal instrument of competition.

The relevant informational practices are not all equivalent. Every campaign must select information and frame its presentation; complete communication is impossible. Selection and framing do not by themselves establish reduced fidelity. Misleading omission, unsupported exaggeration, material ambiguity, deceptive presentation, and intentional falsehood present progressively different questions about whether the resulting communication preserves adequate access to the relevant facts.

A campaign need not begin from a general decision that truth is expendable. It may instead discover that a particular qualification, disclosure, or fidelity constraint is unnecessary to electoral victory under the conditions it confronts. A selective presentation that appears effective can be repeated. A qualification that reduces

persuasive force can be omitted. An opponent's statement can be presented in its least favorable plausible context.

But the absence of a constraint's necessity for victory does not establish the admissibility of removing it. Nor does the effectiveness of a communication establish either its necessity or its admissibility.

Thus:

$E(X, C_a, K)$ does not entail $N(X, C_a, K)$,

and:

$E(X, C_a, K)$ does not entail $A(X, C_a, K)$.

The campaigns also affect one another's conditions. If a communicative technique succeeds for one campaign, the competing campaign may face pressure to answer through a comparable technique, adopt a countermeasure, or accept the electoral cost of restraint. What is initially exceptional can consequently become more available within subsequent competition.

This recursive effect is possible rather than inevitable. Electoral law, journalistic institutions, campaign norms, fact-checking organizations, courts, election officials, and voter responses can preserve constraints or increase the cost of violating them. The relevant claim is not that electoral competition must degrade informational fidelity. It is that locally successful conduct can reconstruct the competitive conditions confronting later conduct.

The campaigns are also not the whole organization. Both compete within a containing electoral organization whose continuation can be designated C_e.

C_e requires more than the production of a winner. It requires sufficiently retained procedures for determining the winner, enough shared evaluability for citizens and institutions to assess competing claims, sufficient legitimacy for authoritative outcomes to remain recognizable, and sufficient expectation of future participation that losing competitors have reason to remain within the electoral process.

The literature on democratic legitimacy commonly treats "loser's consent"—the willingness of those whose preferred candidate loses to accept the authoritative outcome—as an important condition of democratic continuation. Recent scholarship also distinguishes attempts to affect a particular electoral result from disinformation that degrades trust in the electoral process itself.

This produces the central organizational tension. A technique can contribute to C_a or C_b while degrading a condition required for C_e. No single omission, exaggeration, falsehood, or attack need terminate the electoral organization. Repeated relaxations can nevertheless

reconstruct the informational and competitive environment, reducing the cost of further relaxation and increasing the local cost of unilateral restraint.

The resulting claims remain distinguishable:

$E(X, C_a, K)$ does not entail $E(X, C_e, K)$.

$A(X, C_a, K)$ does not entail $A(X, C_e, K)$.

And, more fundamentally:

Effectiveness for C_a does not establish admissibility even for C_a.

Electoral victory and preservation of the containing electoral organization therefore need not favor the same conduct.

The Prince Cross-References

The current seven cross-references make the chapter appear to gather every remotely applicable Machiavellian passage. Four are sufficient.

Chapter XVIII — How princes should keep faith. This is the primary cross-reference. Machiavelli evaluates promise-keeping, appearance, concealment, and deception relative to political effect and *necessità*. The electoral case retains that distinction between informational conduct and local political success while introducing a continuation that Machiavelli does not separately specify: the containing electoral organization.

Chapter VIII — Those who attain a principality through crimes. Machiavelli distinguishes successful acquisition from glory and honor. Acquisition through effective means therefore does not entail success relative to every evaluation of the acquired position. The election case extends this separation: successful acquisition of office does not establish preservation of the institutional organization through which that office was acquired.

Chapter VII — Cesare Borgia and foundations. Machiavelli distinguishes acquisition from the foundations required for later preservation. A campaign may acquire office through conduct that does not supply—and may damage—relations required for legitimate governance after victory.

Chapter XXI — How a prince should act to gain reputation. Campaign communication is partly an organization of public evaluation. Visible commitments, alignments, actions, and distinctions affect how an actor remains publicly recognizable. Modern polling, targeting, media segmentation, and rapid feedback greatly increase the speed and precision with which those evaluations can be observed and acted upon.

The containing electoral organization is a result of the modern application, not a proposition that should be attributed backward to Machiavelli. Machiavelli supplies distinctions among acquisition, preservation, appearance, political effect, and alternative evaluations of success. The election case introduces the additional relation between competing campaign continuations and the continuation of the constitutional system containing their competition.

An AC Weakening

Let A and B be competing campaign organizations.

Let:

C_a = electoral victory by A

and:

C_b = electoral victory by B.

Under the specified election, C_a and C_b are mutually exclusive.

Let C_e designate continuation of the containing electoral organization through which C_a or C_b can become authoritatively established.

A communicative technique X may contribute to C_a:

$E(X, C_a, K)$.

That result does not establish that X is necessary for C_a:

$E(X, C_a, K)$ does not entail $N(X, C_a, K)$.

It also does not establish that X is admissible relative to C_a:

$E(X, C_a, K)$ does not entail $A(X, C_a, K)$.

Even if X were separately established as admissible relative to C_a, that result would not transfer automatically to C_e:

$A(X, C_a, K)$ does not entail $A(X, C_e, K)$.

The competing organizations are recursively coupled. Conduct by A can alter the conditions confronting B, and conduct by B can alter the conditions confronting A. A locally successful relaxation of an informational constraint can therefore make corresponding relaxation more attractive or less avoidable for the competitor.

The cumulative effect can reconstruct K:

$K_1 \rightarrow K_2$.

A technique evaluated under K_1 cannot be assumed to retain the same effectiveness, necessity, or admissibility under K_2.

AC Weakening: Two organizations can pursue mutually incompatible local continuations within a containing organization upon which both depend. A technique that contributes to one local continuation can reconstruct conditions in ways that degrade dependencies required for continuation of the containing organization.

The result must remain carefully bounded. The local effectiveness of X does not establish its local admissibility. And neither local effectiveness nor local admissibility establishes admissibility relative to the containing organization.

This is the crucial indexing move:

"Effective campaign technique" does not entail "admissible campaign technique."

And:

"Admissible for C_a" does not entail "admissible for C_e."

Weaker Still

Remove elections, candidates, parties, electoral information, and political institutions.

Weaker still: Two or more organizations can pursue mutually incompatible local continuations within a containing organization upon whose continuation they depend. A constraint not necessary to a particular local continuation under specified conditions may nevertheless contribute to—or, if separately demonstrated, be necessary for—the continuation of the containing organization.

Competitive advantage can favor relaxation of such a constraint even when repeated relaxation progressively damages conditions required by the containing organization. No competitor need intend the containing damage. Each need only respond instrumentally to its local continuation and to the conditions produced by the other competitors.

What is unnecessary to the continuation of a competing part may nevertheless be necessary to the continuation of the organization containing the competition.

Lincoln: Electoral Decision and Constitutional Continuation

The Lincoln material supplies an unusually sharp historical specification case because Lincoln explicitly distinguishes carrying an election from preserving the political organization through which the result acquires governing authority.

By the time of Lincoln's First Inaugural Address on March 4, 1861, victory in the 1860 election was already complete. The continuation under discussion was no longer:

C_l = Lincoln wins the election.

The operative continuation was:

C_u = continuation of the Union as a constitutional government whose elections remain capable of authoritatively determining office.

Lincoln argues that constitutional government requires acquiescence by one side after an electoral decision:

"If the minority will not acquiesce, the majority must, or the Government must cease."

He does not treat unconstrained majority preference as sufficient. He describes legitimate majority rule as restrained by constitutional checks and limitations and as open to later change through public opinion. Electoral acquiescence is therefore presented as part of a continuing constitutional process rather than permanent submission by the losing side.

Lincoln's July 4, 1861 Message to Congress makes the acquisition/preservation distinction still more explicit. He describes popular government as having succeeded in its establishment and administration but still facing the unresolved problem of maintenance against an internal attempt to overthrow it. He then argues that those capable of fairly carrying an election must also be capable of suppressing rebellion and that an electoral decision cannot be successfully displaced by an appeal from ballots back to force.

This is Lincoln's constitutional argument, not a politically neutral description of the conflict. By July 1861, the relevant conditions included secession, armed rebellion, and civil war. Lincoln's specified continuation was therefore not merely an electoral procedure. It was continuation of the Union and federal constitutional government under conditions in which organized force was being used to reject its authority.

The AC significance lies in the required re-indexing. A measure unnecessary relative to winning the election may be claimed as necessary relative to preserving C_u under the altered conditions K_R:

$N(X, C_l, K_e)$ does not entail $N(X, C_u, K_R)$.

Nor does the absence of necessity under ordinary administrative conditions establish absence of necessity under rebellion:

$N(X, C_u, K_1)$ does not entail $N(X, C_u, K_2)$.

But changing the continuation and conditions does not establish the necessity or admissibility of the measures claimed on their behalf.

Lincoln himself makes the stronger argument that when an end is lawful and obligatory, means indispensable to that end are also lawful and obligatory. AC separates the steps contained in that claim. It requires separate establishment that:

1. the specified end is the continuation actually being preserved;
2. X is genuinely indispensable to that continuation under K;
3. no weaker available intervention preserves the required dependency;
4. X does not unnecessarily damage other dependencies required by the same continuation; and
5. necessity relative to C_u is not silently promoted into admissibility relative to every affected continuation.

Thus:

$N(X, C_u, K_R)$ does not by itself entail $A(X, C_u, K_R)$.

And:

$A(X, C_u, K_R)$ does not entail $A(X, C_2, K_R)$ for every continuation C_2 affected by X.

The First Inaugural makes this final restriction especially important. In the same address, Lincoln disclaims any purpose to interfere with slavery in states where it already exists and discusses the constitutional provision concerning persons "held to service or labor." The constitutional organization he sought to preserve therefore included, at that point, the continued legal availability of slavery within the states where it existed.

Continuation of C_u was consequently not identical to preservation of every person, liberty, or morally relevant continuation contained within or affected by the Union. Establishing a measure as necessary for preservation of constitutional government would not by itself establish that measure—or the constitutional arrangement being preserved—as admissible relative to enslaved persons or every other affected continuation.

Lincoln therefore illustrates both sides of the specification discipline:

- electoral victory does not exhaust continuation of the constitutional organization;
- preservation of the constitutional organization does not exhaust every continuation affected by preserving it.

The case does not establish the admissibility of Lincoln's particular wartime measures. It shows how a completed electoral acquisition can be followed by a distinguishable preservation problem, while leaving every resulting necessity and admissibility claim open to further evaluation.

Analytical Significance

The modern election case extends Machiavelli rather than merely applying his advice to campaign tactics.

The Prince repeatedly distinguishes acquisition of power from preservation of the acquired continuation. A modern constitutional election introduces an additional organizational level: competing campaigns pursue mutually exclusive local continuations through a containing organization whose procedures give victory authoritative effect.

A campaign may not require a particular informational or procedural constraint in order to win under specified conditions. The same constraint may nevertheless contribute to—or, if separately demonstrated, be necessary for—the containing electoral organization through which victory becomes institutionally recognizable.

Lincoln supplies the subsequent distinction. Once the election has been decided, the successful campaign continuation is complete, but continuation of the constitutional organization is not. Acquisition of office and preservation of the electoral-governmental order therefore present different dependency structures.

The principal organizational result is not that candidates lie or manipulate. It is that conduct can remain locally effective or instrumentally intelligible while reconstructing conditions in ways that damage the containing organization upon which the local objective depends.

AC makes the suppressed indices explicit:

$E(X, C_l, K)$ does not entail $E(X, C_o, K)$.

$A(X, C_l, K)$ does not entail $A(X, C_o, K)$.

But AC also prevents the opposite promotion:

Preservation of C_o does not automatically establish the admissibility of every measure claimed as necessary for C_o.

The local and containing evaluations can therefore differ without contradiction. Neither evaluation automatically governs the other.

Source Note

Abraham Lincoln, First Inaugural Address, March 4, 1861. The relevant passages concern constitutional majority rule, minority acquiescence, continuation of government, noninterference with slavery where it existed, and the constitutional provision concerning persons held to service or labor.

Abraham Lincoln, Message to Congress in Special Session, July 4, 1861. The relevant passages distinguish establishment, administration, and maintenance of popular government; connect electoral decision with resistance to rebellion; and state Lincoln's argument concerning lawful ends and indispensable means.

Christopher J. Anderson, André Blais, Shaun Bowler, Todd Donovan, and Ola Listhaug, Losers' Consent: Elections and Democratic Legitimacy (Oxford: Oxford University Press, 2005).

Alexandra Cirone, "Is Democracy Broken? Disinformation Wants You to Think That It Is," in Global Challenges to Democracy: Comparative Perspectives on Backsliding, Autocracy, and Resilience, ed. Valerie J. Bunce, Thomas B. Pepinsky, Rachel Beatty Riedl, and Kenneth M. Roberts (Cambridge: Cambridge University Press, 2025), 215–232, DOI: 10.1017/9781009602570.012.

CHAPTER 3 — CAPACITY BEYOND THE SELECTING CONTINUATION

In Chapter III of *The Prince*, Machiavelli examines Louis XII's intervention in Italy.

Louis entered Italy from a position of considerable advantage. He acquired Milan quickly, and numerous lesser Italian powers initially sought association with France. Yet Louis subsequently altered the political organization within which that advantage had become available.

Among his actions, he assisted Pope Alexander VI and Cesare Borgia in their expansion into Romagna. Louis's assistance served identifiable purposes of his own. Machiavelli notes the promise to assist the pope in return for the annulment of Louis's marriage and a cardinal's hat for the Archbishop of Rouen. Whatever its immediate transactional value, the assistance also increased the temporal capacity of the papacy and of Cesare Borgia's organization.

The capacity thereby produced did not remain confined to the purposes for which Louis supplied it.

Alexander and Cesare possessed continuations of their own. Increased papal and Borgia capacity remained available to those continuations after the transaction that had made Louis willing to contribute to it.

Machiavelli places this decision among a larger series of errors. Louis weakened lesser powers that might have remained dependent upon him, strengthened the Church, introduced Spain as another great foreign power in Italy, failed to reside there, and eventually moved against Venice.

His conclusion is severe:

Whoever causes another to become powerful causes his own ruin.

The immediate claim is political and stronger than the historical sequence alone establishes as a universal rule. Machiavelli's example nevertheless supplies a more limited organizational relation. An actor can increase another organization's capacity because that increase is expected to serve the actor's own continuation. The other organization retains its own continuation, however, and the increased capacity remains available beyond the purpose that caused the actor to produce it.

The relevant political relation is therefore not merely:

Louis helped another power.

It is:

Louis selected an intervention for purposes of his own; the intervention increased capacity retained by another organization; and that capacity remained available to a continuation not controlled by Louis.

An AC Weakening

Remove Louis XII, Alexander VI, Cesare Borgia, France, the papacy, and Italy.

Let A and B be organizations possessing distinguishable continuations C_a and C_b.

A selects intervention X because an increase in B's capacity is expected to contribute to C_a.

The selection can be represented schematically:

X contributes to ΔB^+, and ΔB^+ is expected to contribute to C_a.

This representation records A's reason for selecting X. It does not establish that X was actually effective for C_a, still less that X was necessary:

Selection of X for C_a does not establish $E(X, C_a, K)$.

$E(X, C_a, K)$ would not by itself establish $N(X, C_a, K)$.

B nevertheless possesses its own continuation, C_b. Because the increased capacity is retained by B, its availability is not restricted to the continuation that supplied A's reason for producing it.

Thus:

ΔB^+ can remain available to C_b.

If subsequent conditions make C_b incompatible with C_a, the capacity A helped produce can contribute to consequences adverse to A.

Machiavelli's political observation can therefore be weakened:

An intervention that increases another organization's capacity because that increase is expected to contribute to one's own continuation does not thereby restrict the increased capacity to that continuation.

The intervention is selected relative to C_a.

The increased capacity remains organized through B.

This formulation preserves the source relation:

- distinguishable organizations;
- distinguishable continuations;
- an intervention selected for one continuation;
- increased capacity retained by another organization;
- and possible use of that capacity beyond the selecting continuation.

It does not require the universal Machiavellian conclusion that whoever increases another's power necessarily causes his own ruin.

Weaker Still

The preceding formulation retains A's intention to increase B's capacity.

That intention can be removed if the object of analysis is limited to organizational consequence rather than Louis's prudence, knowledge, or responsibility.

Suppose X is selected for C_a but alters a condition K that is also accessible to B. If the alteration increases capacity available to C_b, that consequence does not depend upon A having intended to strengthen B.

What remains is:

An intervention selected for C_a can alter a shared condition in ways that increase capacity available to C_b, whether or not increasing that capacity was A's purpose.

A further epistemic condition can then be examined. Must A know that B exists?

The causal dependency does not require that knowledge. If B has access to the altered condition, B can be affected whether or not B was represented in A's selection specification.

Removing A's knowledge therefore preserves the organizational consequence, but it does not preserve every feature of Machiavelli's political judgment. Louis knowingly dealt with Alexander and Cesare; the resulting criticism concerns his statecraft. An actor unaware of B may instantiate the weakened causal relation without instantiating Louis's imprudence.

The more limited proposition is:

A continuation absent from the specification under which an intervention is selected can nevertheless be affected when the intervention alters a condition accessible to that continuation.

Freeze the following bounded proposition for cross-domain evaluation:

An intervention selected for one continuation can produce consequences for another continuation when it alters conditions accessible to both.

Selection relative to one continuation does not itself restrict the intervention's consequences to that continuation. But cross-continuation consequence is not automatic. It requires a relation through which another continuation can access or be affected by the altered condition.

The proposition can therefore fail to instantiate where:

- the altered condition is not shared;
- no other continuation has access to or depends upon it; or
- the intervention is sufficiently isolated that its consequences remain confined to the specified continuation.

A Cross-Domain Thought Experiment

Consider a handful of different seeds planted in nutrient-poor, sandy soil.

Twelve of the seeds are known to be sweet peas. Other seeds are present, but their identities and requirements are unknown.

Let:

C_{sp} = continued flourishing of the sweet peas over the specified horizon.

The articulated condition set includes the sandy soil, limited nutrients, inadequate water, the identified sweet peas, and the presence—but not the identity or continuational requirements—of the other seeds:

$K_{articulated}$.

A gardener selects a composite intervention X consisting of fertilizer and water because those additions are expected to improve conditions for C_{sp}.

The thought experiment stipulates that the intervention is effective in improving the growth of the sweet peas:

$E(X, C_{sp}, K_{articulated})$.

It does not establish that X is necessary for C_{sp}. Nor does it separately establish the contribution of fertilizer and water considered independently.

The intervention works in the stipulated sense. Nutrients and water become more available, and the sweet peas grow more successfully.

Then the unidentified seeds germinate.

They are thistles.

Nothing in the stipulated result requires the fertilizer or water to have failed relative to the sweet peas. The incompleteness lies in the articulated relational environment.

We fertilized **for the sweet peas**.

But we fertilized **the soil**.

The nutrients and water are not restricted by the continuation that supplied the gardener's reason for adding them. Once introduced into the shared soil, they become available to whatever organisms are capable of exploiting them.

The thistles therefore benefit from an intervention selected relative to C_{sp} even though continuation of the thistles was absent from the gardener's specification.

The actual relational environment includes sweet peas, thistles, shared soil, shared nutrients, shared water, and competition for those resources.

Thus:

$K_{actual} \neq K_{articulated}$.

The original effectiveness claim need not have been false. The intervention may genuinely improve sweet-pea growth.

But:

$E(X, C_{sp}, K_{articulated})$

does not establish:

$E(X, C_{sp}, K_{actual})$.

If the thistles exploit the improved conditions more aggressively, the intervention can improve the immediate growth of the sweet peas while reducing their longer-term continuability within the shared plot.

The continuation was never realized in isolation. It was always:

C_{sp} under K_{actual}.

The frozen proposition instantiates:

An intervention selected for one continuation can produce consequences for another continuation when it alters conditions accessible to both.

The fertilizer and water were selected for C_{sp}. They altered shared conditions. Those changed conditions became available to continuation of the thistles.

Because the example was deliberately constructed around shared soil and unidentified seeds, this instantiation should not be treated as independent confirmation. Its purpose is analytical: to determine whether the weakened relation remains coherent and informative after political power, intentional strengthening of another actor, and knowledge of the affected continuation have been removed.

It does.

What the Garden Adds

The political and garden cases must now be separated by provenance.

Louis knew that Alexander VI and Cesare Borgia existed. He knowingly contributed to their capacity. Machiavelli can therefore criticize Louis's judgment.

The gardener knows that other seeds exist but does not know that they are thistles. Their continuations are not incorrectly evaluated. They are absent from the operative specification.

The garden consequently exposes a further result:

Unspecified continuations are not thereby unaffected continuations.

It also exposes a related specification problem:

An intervention evaluated for a continuation under an incompletely articulated condition set cannot be assumed to preserve that continuation within the fuller relational environment affected by the intervention.

These results belong initially to the garden thought experiment and the AC reconstruction. They should not be attributed backward to Machiavelli without separate textual support.

The provenance sequence is:

Louis XII → capacity retained by another organization remains available beyond Louis's purpose → intentional strengthening removed while shared accessibility is retained → knowledge of the affected continuation removed → frozen proposition → sweet peas and thistles → unspecified continuations remain affectable.

Each transition must be classified correctly.

The movement from Louis to the frozen proposition is AC Weakening.

The movement from the frozen proposition to the garden is cross-domain instantiation.

The proposition concerning unspecified continuations is a downstream result generated by the garden analysis.

The garden therefore does not show that Machiavelli described ecology, unintended consequences generally, or a universal law of intervention. It shows something narrower:

A relation reconstructed from Machiavelli's political analysis remains coherent after political identity, political power, intentional strengthening, and knowledge of the affected continuation have been removed.

Because the garden is a constructed thought experiment, that survival demonstrates analytical portability rather than independent empirical confirmation.

The remaining question is how far the relation can be transported while retaining the shared-access condition that prevents it from becoming vacuous.

CHAPTER 4 — INJURY, PRESERVATION, AND THYROID CANCER

Machiavelli examines injury not only at the point where it is imposed but relative to the political continuation within which its consequences propagate.

In Chapter XVII, this distinction appears in his comparison between limited exemplary punishment and disorder allowed to continue. The punishments injure particular persons. The murders and robberies arising from continued disorder, he argues, injure the wider community. His comparison is consequential and politically indexed: conduct conventionally described as merciful can permit more widely distributed injury, while conduct described as cruel can contribute to the continuation he has specified.

Chapter VIII adds a temporal restriction. Machiavelli distinguishes cruelties "well used" from cruelties badly used. The former are concentrated at the beginning, employed for security, not continually repeated, and followed by measures useful to subjects. The latter increase rather than diminish.

This is Machiavelli's instrumental classification, not an AC determination of moral or organizational admissibility. It nevertheless supplies a temporal relation relevant to AC: an intervention that contributed to preservation under one condition set cannot be assumed to remain effective, necessary, or warranted when its duration, intensity, target, or surrounding conditions change.

Chapter IX contributes another distinction. The people and the great occupy different relations within the political organization. Machiavelli argues that they cannot be removed, replaced, alienated, or controlled with equivalent consequences. The organizational position of an affected component therefore matters to the continuation being analyzed.

These chapters do not expressly state one unified proposition. WMW reconstructs a relation across them:

The consequence of an injurious intervention depends upon what it damages, what threatening process it opposes, how injury propagates, how long the intervention continues, and which continuation is being evaluated.

This is more precise than the familiar claim that Machiavelli permits harmful means for beneficial ends. Harm may arise from intervention, nonintervention, or available alternatives. The presence of harm alone does not complete the evaluation.

An AC Weakening

Let O be an organization whose specified continuation is C_o.

Let P be a process operating within O such that, under conditions K_1, continued operation of P threatens a dependency required for C_o.

Let X be an intervention that inhibits P but also damages some components or relations belonging to O.

The fact that X produces damage does not by itself establish that X is inadmissible:

Damage produced by X does not entail not-A(X, C_o, K_1).

But neither does preservation establish admissibility:

E(X, C_o, K_1) does not entail N(X, C_o, K_1).

N(X, C_o, K_1) does not entail A(X, C_o, K_1).

The relevant comparison is not simply:

damage versus no damage.

It is among the organizational consequences of:

- intervention X;
- withholding X;
- weaker or differently targeted interventions;
- and any other available alternatives.

Those consequences cannot necessarily be reduced to an aggregate quantity. Damage to components occupying different dependency relations can have different consequences for C_o even when their magnitude appears similar.

The evaluation must also preserve the actual intervention. Necessity or admissibility established for one target, intensity, duration, or distribution does not automatically transfer to another:

N(X_1, C_o, K_1) does not entail N(X_2, C_o, K_1).

And necessity under one condition set does not transfer automatically to another:

N(X, C_o, K_1) does not entail N(X, C_o, K_2).

The same restriction applies to admissibility:

A(X, C_o, K_1) does not entail A(X, C_o, K_2).

Machiavelli's princes, subjects, nobles, punishments, cruelty, reputation, and political disorder can now be removed. What remains is an organizational relation in which a process internal to an organization threatens the organization's continuation, while an intervention against that process also damages contributing parts or relations.

The intervention-caused damage remains part of the evaluation. It is not cancelled by preservation of C_o.

Weaker Still

An organization need not preserve every component or relation unchanged in order to continue.

Under some conditions, a process operating within an organization may threaten its continuation. Inhibiting that process may require, or may be attempted through, an intervention that also damages components or relations belonging to the same organization.

The existence of intervention-caused damage does not by itself determine admissibility. Neither does the intervention's effectiveness.

Freeze the following propositions:

F_1. An intervention can damage components of an organization while contributing to continuation of the organization containing them.

F_2. The effectiveness, necessity, and admissibility of an intervention remain indexed to its continuation, conditions, target, extent, and duration.

F_3. The organizational consequence of damage depends upon the relations occupied by what is damaged, not merely upon an aggregate amount of damage.

These propositions do not grant a general license to damage parts. F_1 establishes only compatibility between local damage and containing preservation. F_2 and F_3 preserve the conditions required for evaluating whether a particular intervention is effective, necessary, or admissible.

Nothing specifically political remains in them.

A Cross-Domain Case Study: Thyroid Cancer

Consider treatment of thyroid cancer that has extended beyond the thyroid and involved surrounding tissue.

The malignant process is not an external attacker. It originated from cells belonging to the organism. Its continued proliferation can nevertheless become incompatible with continuation of the patient.

Let:

C_p = continuation of the patient

and:

C_m = continued proliferation of the malignant process.

Under conditions in which continued malignant proliferation threatens C_p:

C_m contributes to reduced continuability of C_p.

An intervention X_t may be selected to eliminate or sufficiently suppress C_m. But X_t operates within the same organism whose continuation it is intended to preserve.

In the history considered here, treatment was composite and included surgery, radioactive iodine, and external-beam neck irradiation. The result therefore belongs to the composite treatment unless the contributions of its components are separately evaluated.

The treatment was followed by long-term cancer control and survival. It also produced or was followed by lasting injury within the treated region. Throat and neck injury remained part of the patient's subsequent condition set. Carotid stenosis later developed and required stenting.

The published medical literature independently supports the broader association between therapeutic neck irradiation and delayed vascular injury, carotid stenosis, and cerebrovascular risk. It also documents carotid stenting and endarterectomy as treatments used for radiation-induced carotid stenosis.

That literature supports the general clinical relation:

external-beam neck irradiation can contribute to delayed carotid vascular injury.

It does not independently document this patient's history, establish the cause of every later injury in this individual, or attribute carotid injury to radioactive iodine. The personal history remains firsthand testimony unless supported by the patient's clinical records.

The result is not contradictory:

The treatment damaged parts of the organism.

And:

The treatment contributed to preservation of the organism.

Both can be true because the intervention affected healthy contributing structures while opposing a process that more seriously threatened C_p.

The case therefore instantiates F_1:

An intervention can damage components of an organization while contributing to continuation of the organization containing them.

This does not by itself establish that every component of X_t was necessary or that the composite intervention was admissible. Those conclusions would require the patient's complete clinical conditions, available alternatives, expected outcomes, target distributions, doses, and relevant horizon.

The case also instantiates the condition-indexing requirement in F_2. If a particular intervention was necessary under K_1, while the malignancy presented a specified threat, that necessity does not automatically transfer to a later condition set:

$N(X_t, C_p, K_1)$ does not entail $N(X_t, C_p, K_2)$.

This does not imply that treatment should cease as soon as malignant tissue is reduced or no longer detectable. Adjuvant or continuing treatment may remain necessary under the fuller clinical condition set. The AC claim is only that the earlier necessity result cannot be transferred without renewed evaluation.

Likewise:

$N(X_t, C_p, K_1)$ does not entail $N(X_t + \Delta X, C_p, K_1)$.

Necessity at one intensity or duration does not establish necessity at a greater intensity or duration.

The case also instantiates F_3. Malignant and healthy tissues are not organizationally equivalent merely because both occur within the same organism. Continued malignant proliferation threatens C_p. Healthy structures ordinarily contribute to C_p. Damage to them therefore has different consequences relative to the patient's continuation.

This does not mean that every malignant cell can be destroyed without cost or that every healthy structure must be preserved unchanged. It means that organizational position and contribution matter to the evaluation.

What the Case Adds

The thyroid-cancer history exposes a temporal consequence not required by the Machiavellian derivation.

An intervention X_t applied under K_1 can contribute to C_p while also reconstructing the conditions under which C_p must subsequently continue:

Application of X_t under K_1 contributes to K_2.

Some elements of K_2 may later constrain C_p and require further intervention.

Thus:

An intervention can contribute to preservation of a continuation under one condition set while producing damage that becomes a constraint upon that same continuation under a later condition set.

This proposition avoids claiming that the original intervention was admissible. It identifies the temporal organization independently of that unresolved judgment.

The later damage does not establish that the original intervention was unnecessary or inadmissible. Nor does successful cancer control make the later damage irrelevant. Both belong to the organizational history of C_p.

This result belongs initially to the thyroid-cancer case. It should not be attributed backward to Machiavelli without separate textual support.

Methodological Result

The path is:

Chapters XVII, VIII, and IX → reconstructed relation among localized injury, containing continuation, temporal limitation, and organizational position → political particulars removed → frozen propositions → thyroid-cancer case study → intervention reconstructs later conditions.

This is not a direct analogy between Machiavellian cruelty and cancer treatment.

Nor is it independent empirical confirmation of a Machiavellian invariant. The cancer case was deliberately selected because it contains the relevant intervention structure.

Its contribution is more limited and defensible:

Relations reconstructed from Machiavelli's analyses of injury and political preservation remain coherent and informative in a biological organization containing no prince, subjects, political authority, culpability, or intentional adversary.

Cancer need not intend to destroy the organism. Malignant cells need possess no political or psychological objective. Their continued proliferation need only become incompatible with continuation of the organism containing them.

The case therefore functions as an agency-removal application and as a source of a downstream temporal result.

Source Note

Ping-Yi Lin et al., “Risk of CVD Following Radiotherapy for Head and Neck Cancer: An Updated Systematic Review and Meta-Analysis,” *Frontiers in Oncology* 12 (2022): 820808. DOI: 10.3389/fonc.2022.820808. This review reports increased cerebrovascular and carotid-stenosis risk following head-and-neck radiotherapy.

Jiaping Xu and Yongjun Cao, “Radiation-Induced Carotid Artery Stenosis: A Comprehensive Review of the Literature,” *Interventional Neurology* 2, no. 4 (2014): 183–192. DOI: 10.1159/000363068.

Jihee Kang et al., “Treatment Results of Carotid Endarterectomy and Carotid Artery Stenting for Patients with Radiation-Induced Carotid Stenosis,” *Annals of Surgical Treatment and Research* 103, no. 2 (2022): 112–118. DOI: 10.4174/astr.2022.103.2.112.

CHAPTER 5 — CCIBASIC: REDISTRIBUTING CAPABILITY WITHOUT REDISTRIBUTING EXPERTISE

Machiavelli repeatedly distinguishes receiving useful effects from forces supplied by others and possessing foundations through which the required force remains available under one’s own control.

In Chapters XII and XIII of *The Prince*, this distinction appears most clearly in his treatment of mercenary and auxiliary arms. A prince may receive military effects from forces he does not command as his own. Those forces may even be competent and successful. Their effectiveness does not, however, establish that the capability will remain available under conditions determined by the prince’s continuation.

Auxiliary arms make the problem especially clear because they remain organized under another ruler. If they fail, the prince is defeated; if they succeed, the prince may become dependent upon the power whose forces produced the victory. The immediate military objective can therefore be achieved without securing the continued availability of the capability that achieved it.

Chapter VII presents a related problem through Cesare Borgia. Borgia initially acquires power through conditions substantially supplied by the position and fortune of his father, Pope Alexander VI. Machiavelli then examines Borgia’s attempt to construct foundations capable of surviving the disappearance of those enabling conditions. The problem is not that externally supplied forces cannot produce results. It is that results obtained through them do not by themselves remove the dependency through which those results became possible.

The derived distinction is therefore not simply between effective and ineffective means.

It is between using an effective capability and controlling or preserving the relation through which that capability remains available.

An AC Weakening

Remove princes, armies, mercenaries, auxiliary forces, conquest, and political power.

Let organization A possess a specified continuation, C_a, for which capability Q is required under conditions K.

Under the existing arrangement, A obtains Q through a relation with organization B:

$$\mathbf{R_{ab} \rightarrow Q_a \rightarrow C_a}$$

The arrows represent an access or dependency relation, not logical implication. R_{ab} is the relation through which B makes Q available to A, and Q_a is Q as it becomes available for A's continuation.

Q may be entirely effective when supplied. Nevertheless, successful use of Q does not establish that Q will remain available if R_{ab} changes or disappears.

Nor does the present availability of Q establish that B itself is necessary for C_a. B might be replaceable, A might reconstruct access to Q, or another capability might serve the same dependency. What has been established is narrower: under the existing arrangement, access to Q remains dependent upon a relation whose continuation is not controlled solely by A.

The Machiavellian relation can therefore be weakened as follows:

An organization may make effective use of a capability without controlling the relation upon which the capability's continued availability depends.

And more specifically:

Where a continuation depends upon continued access to a capability through a relation the organization cannot preserve or replace, successful prior use of that capability does not remove the resulting vulnerability.

This is the weakest proposition required for the present evaluation. Freeze it here.

A Cross-Domain Engineering Case Study

Consider a prototype-production environment in which automated test programs must be created for electronic products moving from engineering into production.

The test equipment exists. Product engineers understand the products, know what must be measured, and can interpret the resulting measurements. Creation of executable automated tests nevertheless depends upon a specialized capability concentrated in one engineer: the programming and instrument-control expertise required to convert the testing requirements into working software.

Let:

- H_e designate the specialist's originating expertise;
- Q_t designate the bounded capability to create ordinary automated product-test programs;
- C_t designate continued organizational ability to create those programs.

Under the initial arrangement:

$$\mathbf{H_e \rightarrow Q_t \rightarrow C_t}$$

The problem is not that the specialist's capability is ineffective. It is effective precisely because the specialist can create the required programs. The vulnerability lies in the access relation. Every new automated test requiring that implementation capability must return through the same engineer.

Product engineers possess information required by the testing process, but they cannot independently convert that information into executable test programs. The organization therefore possesses the required capability only through a concentrated relation.

CCIBASIC was created to reconstruct that relation.

CCIBASIC provided a BASIC-like interpreted language through which engineers without the underlying C++ and instrument-control expertise could construct bounded automated test programs. It embodied selected consequences of the specialist's expertise in an executable artifact and exposed them through an interface appropriate to the engineers who needed to use them.

Let I designate the interpreter. The reconstructed relation can be represented as:

$$\mathbf{H_e \rightarrow I}$$

and:

$$\mathbf{I + product\ knowledge \rightarrow Q_t \rightarrow C_t}$$

CCIBASIC did not transfer all of H_e to the product engineers. Those engineers did not thereby acquire the ability to recreate the interpreter, implement its instrument-control layer, or solve every programming problem that the specialist could solve.

What changed was narrower and organizationally significant: creation of each ordinary test program no longer required the specialist's direct implementation.

The frozen proposition is therefore instantiated, assuming the accuracy of the organizational history supplied by the author:

An organization may make effective use of a capability without controlling the relation upon which the capability's continued availability depends.

Before CCIBASIC, the organization obtained effective automated tests, but access to the capability required for each new test remained concentrated in one recurring relation. CCIBASIC altered that relation by preserving a bounded realization of the capability in an artifact usable by additional participants.

The case does not show that every dependency was eliminated. It shows that recurring access to a specified capability became less concentrated.

What the CCIBASIC Case Adds

The case produces a result not contained in the frozen proposition:

A bounded capability can be made available to additional participants without equivalently redistributing the expertise from which that capability was constructed.

Expertise and capability are therefore distinguishable.

The specialist possessed expertise sufficient to create and maintain the interpreter, implement instrument communication, and construct test programs directly. Users of CCIBASIC acquired something narrower: access to selected executable consequences of that expertise.

A concentrated-expertise problem therefore need not always be addressed by reproducing the complete expertise in additional people. An organization may instead embody part of its practical yield in an artifact, procedure, interface, or institution through which a bounded capability becomes available to others.

This is a result generated by the CCIBASIC case. It should not be attributed backward to Machiavelli without a separate textual derivation.

The Remaining Dependency

The reconstruction must also be indexed to its continuation.

For C_t—the continued ability of product engineers to create ordinary automated test programs—CCIBASIC reduced recurring dependence upon the specialist's direct participation.

But consider a different continuation:

C_i = continued ability to modify, repair, or extend the interpreter itself.

Relative to C_i, the dependency structure may remain substantially concentrated:

$$\mathbf{H_e \rightarrow C_i}$$

CCIBASIC can therefore reduce dependence upon the specialist relative to C_t while leaving substantial dependence upon the same specialist relative to C_i.

Thus:

Reduced dependency relative to C_t does not entail reduced dependency relative to C_i.

The dependency was not eliminated absolutely. It was reconstructed relative to a particular continuation and horizon.

This produces a second downstream result:

An intervention can reduce a dependency relative to one continuation while preserving that dependency relative to another.

The proper description is therefore not that CCIBASIC eliminated a single point of failure or made the organization independent of software expertise. It reduced recurring dependence for the creation of a specified class of test programs while leaving other dependencies—including interpreter maintenance—potentially in place.

Analytical Significance

Machiavelli supplies the source relation: a capability received through another organization can be effective while leaving the recipient dependent upon the relation through which that capability remains available.

The political particulars can then be removed, producing the frozen proposition:

Successful use of a capability does not by itself establish control over, or continued availability of, that capability.

CCIBASIC provides a cross-domain case in which that relation remains evaluable. It also contributes two further findings:

1. A bounded capability can be redistributed without reproducing all the expertise from which it was constructed.
2. Dependency can be reduced relative to one continuation while remaining relative to another.

The inferential path is therefore:

Machiavelli's own and others' arms → effective capability can remain relationally dependent → frozen dependency proposition → CCIBASIC case → artifact-mediated redistribution and continuation-specific residual dependency.

That does not make CCIBASIC Machiavellian. Nor does it establish a universal law of organizational capability.

It shows that a dependency relation derived from Machiavelli remains analytically useful in an engineering organization radically unlike the political organizations through which he observed it. The engineering case then exposes additional relations that belong initially to that case and must be tested separately if they are to travel further.

Source Note

CCIBASIC is a surviving software artifact written by the author for automated electronic-product testing at Coilcraft, Inc., in the late 1990s. The surviving C++ source can document architectural features visible in the artifact: that it loads and interprets user-supplied programs, supplies BASIC-like control structures, and exposes instrument-communication, measurement, statistical, and hardware-control functions through a bounded command interface.

The source code alone cannot establish the organizational circumstances of the program's creation, the concentration of the preceding implementation capability, its actual adoption by other engineers, or the degree to which it altered the production-test workflow. Those claims rest upon the author's firsthand recollection and should be identified as firsthand historical testimony rather than independently documented organizational history.

CHAPTER 6 — INCH-BASED AND METRIC STANDARDS

In Chapter VI of *The Prince*, Machiavelli identifies the introduction of new orders as one of the most difficult undertakings available to a political innovator.

The difficulty does not arise merely because a new order is inferior, unfamiliar, or technically defective. Machiavelli directs attention to relations already organized around the existing order. Those who benefit from the old arrangements become active opponents of the innovation, while those who might benefit from the new arrangements

initially support it only weakly. The established order therefore possesses an advantage that cannot be reduced to its intrinsic merits.

The old order is already instantiated.

People occupy positions within it. Expectations have formed around it. Advantages depend upon it. Practices reproduce it. The proposed replacement, by contrast, initially exists largely as a possibility whose benefits have not yet been experienced.

These are organizational reconstructions of Machiavelli's political account rather than terms he explicitly supplies. Their shared claim is that an established order is not merely one option considered alongside another. It is an option already connected to interests, practices, expectations, and capacities that contribute to its continuation.

The actual comparison is therefore not simply between:

O_1

and:

O_2.

It is between:

O_2

and:

O_1

together with the organization already constructed through O_1.

This asymmetry can persist even when O_2 possesses properties that would have made it preferable had the two alternatives begun under otherwise equivalent conditions.

An AC Weakening

Remove princes, political factions, conquest, and political innovation.

Let O_1 be an established organizational structure and O_2 a possible replacement. Suppose both could support a specified continuation C under appropriate conditions.

If the comparison were made before either structure had been instantiated, O_2 might be equally or more effective relative to C.

But O_1 is not merely an abstract alternative. Under conditions K_1, components and relations may already have become dependent upon, coordinated through, or compatible with O_1:

$$\mathbf{O_1 \rightarrow \{R_1, R_2, R_3, \ldots R_n\} \rightarrow C}$$

The arrows represent organizational dependency or coordination, not logical implication.

Replacing O_1 therefore may require more than establishing O_2. Let R* designate the subset of required relations organized through O_1 that cannot continue unchanged under O_2. Replacement of O_1 may then require some combination of:

- reconstructing members of R*;
- replacing them;
- maintaining duplicate relations;
- introducing conversions or interfaces between O_1 and O_2;
- discontinuing continuations that remain dependent upon O_1.

The difficulty of replacement therefore need not arise from an intrinsic defect in O_2. It may arise from the organization accumulated through continued use of O_1.

Thus:

The comparative admissibility of an established structure and a proposed replacement cannot be determined solely from their intrinsic properties when required relations have already become organized through the established structure.

And:

Continued use of a structure can create or preserve additional dependencies that must later be addressed if the structure is replaced.

This is more specific than saying that organizations resist change.

The established structure need possess no intention to resist anything. Participants need not regard it as intrinsically superior. No central authority need prohibit replacement.

The structure's continuation advantage may reside in the live dependencies, compatibilities, and practices accumulated around it.

Weaker Still

The preceding formulation still refers to complete organizational orders.

Remove that requirement.

O_1 and O_2 may be standards, conventions, interfaces, protocols, representations, procedures, classifications, units, or other relational structures through which activity is coordinated.

The existing structure need only have become sufficiently instantiated that some relations required for a specified continuation depend upon its continued availability or cannot operate unchanged through its replacement.

What remains is:

A structure can become more costly to replace as required relations become organized through it, independently of whether the structure itself becomes more effective.

“Continued instantiation can produce its own preservation conditions” is weaker, but it approaches vacuity. Almost any repeated practice can be said to acquire some persistence.

The more informative proposition is:

The replacement cost of an instantiated structure depends not only upon the structure being replaced, but also upon the required relations that must be reconstructed, replaced, duplicated, or interfaced because they have become organized through it.

The proposition has a boundary. If no required relations depend upon O_1, or if all such relations remain fully compatible with O_2, the predicted reconstruction cost may be negligible.

Freeze the bounded proposition here.

A Cross-Domain Engineering Evaluation

Consider an engineering and manufacturing organization whose existing products were developed using inch-based dimensions, components, and standards.

The question is not whether inches are intrinsically superior to millimetres. Suppose the organization decides that SI-based engineering would be preferable for future designs.

If the problem consisted solely of changing the unit in which a length is expressed, numerical conversion would be straightforward:

1 in = 25.4 mm

But an engineering standard is not instantiated only in numerical expressions.

Dimensions appear in drawings. Fasteners possess standardized diameters, pitches, thread forms, and mechanical requirements. Tools correspond to those fasteners. Material stock

comes in established sizes. Tolerances have been developed around particular designs and processes. Gauges, fixtures, test equipment, documentation, software, supplier catalogs, inventory systems, purchasing practices, employee knowledge, and customer interfaces may all contain relations organized around the existing standards.

The organization therefore does not face only:

inch-based measurement versus metric measurement.

It faces:

metric standards

versus:

inch-based standards

together with the live organization already instantiated around them.

The frozen proposition predicts that the difficulty of replacement will depend not merely upon the ease of expressing existing dimensions metrically, but upon the number and significance of required relations that cannot continue unchanged under the replacement.

The distinction between numerical conversion and organizational replacement can be seen in a simple fastener example.

A nominal 1/4-inch dimension can be expressed as 6.35 millimetres. But describing an existing 1/4-inch fastener as 6.35 millimetres does not transform it into an M6 fastener. The nominal diameter, thread series and pitch, mating component, tooling, procurement designation, and applicable standards remain those of the original fastener.

Replacement therefore requires decisions about which existing relations will be:

- retained;
- converted only in description;
- redesigned around metric components;
- duplicated during transition;
- connected through interfaces;
- discontinued with the products that require them.

Official U.S. metrication studies have documented precisely this broader transition problem, including revisions to engineering drawings, manuals, standards, catalogs, inventories, equipment, and personnel practices. They also document phased transition and periods of dual-unit or dual-standard support. therefore provides external support for the organizational structure of the case.

The case instantiates the frozen proposition: the difficulty of replacement does not reside solely in changing the unit used to express a measurement. It also resides in the relations organized around the established standard.

The Transitional Organization

An attempted transition from one standard to another can produce an intermediate organization more complex than either endpoint:

$$\mathbf{O_1 \rightarrow \{O_1, O_2\} \rightarrow O_2}$$

The intermediate state is not guaranteed, and it need not occur in every conversion. But it may become necessary when existing products, components, customers, suppliers, or records continue to require O_1 while new work is organized through O_2.

The transition can therefore require temporary preservation of the structure being replaced.

A factory cannot necessarily stop recognizing every inch-based drawing, tool, part, fixture, and product merely because new engineering work becomes metric on a specified date. Existing continuations may require the former standard while new continuations are established through the replacement.

In some organizations, the final transition to O_2 alone may never become admissible. Live products or external interfaces may continue to require O_1 indefinitely.

This produces a downstream result not contained in the frozen proposition:

Establishing a replacement can require temporary—or continuing—preservation of the structure being replaced.

And:

Successful adoption of a replacement does not by itself establish the admissibility of removing the replaced structure.

Continuation-Indexed Coexistence

Let:

- C_n = continued development of new metric designs;
- C_l = continued support of legacy inch-based products.

Under specified conditions, O_1 may remain necessary for C_l while not being necessary for C_n:

$N(O_1,C_l,K)$ may hold while $\neg N(O_1,C_n,K)$.

At the same time:

A(adopt O_2,C_n,K) does not entail A(remove O_1,C_l,K).

Even the first necessity claim must remain open to alternatives. O_1 is not necessary for C_l if legacy support can be preserved through admissible conversion, redesign, replacement, or interfacing. The necessity claim applies only where removing O_1 would eliminate a dependency required for C_l under the specified conditions and horizon.

The coexistence of O_1 and O_2 therefore need not be evidence of irrational resistance to change. It may reflect distinct live continuations whose dependencies have not yet been consolidated under one standard.

This is continuation-indexed coexistence.

Analytical Significance

The weak analogy would be:

Machiavelli says new orders encounter resistance; the United States has resisted metrication.

That comparison establishes very little.

The stronger derivation is:

Machiavelli's account of established orders → existing relations advantage the instantiated order → removal of politics and intentional opposition → replacement difficulty resides partly in dependent relations → frozen proposition → inch-based and metric engineering standards → transitional coexistence → adoption of a replacement does not establish admissibility of removing the replaced structure.

The engineering evaluation therefore contributes something that should not be attributed backward to Machiavelli:

A replacement can be successfully instantiated while the structure it replaces remains necessary relative to another continuation.

The result also clarifies the word "legacy."

A legacy structure is not necessarily an obsolete structure. "Legacy" identifies its historical relation to a newer structure. Whether it is obsolete depends upon whether any live continuation still requires it—or whether those dependencies can be reconstructed through an admissible alternative.

CHAPTER 7 — FORMAT COMPETITION: VHS/BETAMAX AND BLU-RAY/HD DVD

In Chapter III of *The Prince*, Machiavelli analyzes political power as something produced not only by the intrinsic capacities of a prince but also by the surrounding distribution of allies, dependents, rivals, and lesser powers.

When Louis XII entered Italy, Machiavelli's criticism was not simply that Louis made several individually poor decisions. Louis reconstructed the political field within which his own position had to continue. He weakened some powers, strengthened others, admitted another major power, and altered the relations upon which his position depended.

The resulting lesson is broader than the warning that one should not make another powerful.

A political actor's effective strength cannot always be determined by examining that actor in isolation. Other organizations may supply resources, constrain competitors, provide access, confer legitimacy, or organize their own conduct around that actor. Those relations become part of the conditions under which the actor's continuation is evaluated.

Chapter IX provides limited support for the same reconstruction. A prince elevated by the people occupies a different organizational position from one elevated by the great, even if the nominal office is the same. The supporting relations differ, and those differences affect what the prince can do and what threatens his continuation.

These chapters support a relational proposition:

An organization's effective strength can be partly produced by the relations organized around it.

They do not yet establish a theory of technological adoption, network effects, or path dependence.

An AC Weakening

Remove princes, states, political alliances, conquest, and political power.

Let A and B be alternative structures capable of contributing to a specified continuation C_i under conditions K, where C_i is the continuation of the organization choosing between them.

Let R_A and R_B designate the surrounding relations respectively coordinated through A and B.

These relations may include complementary capabilities, compatible artifacts, infrastructure, suppliers, users, accumulated knowledge, distribution channels, and expectations concerning future availability.

The effective contribution of A to C_i need not depend solely upon properties internal to A. It may also depend upon R_A.

Accordingly, two alternatives cannot always be evaluated as isolated objects:

A

versus:

B.

The relevant comparison may instead be:

A together with R_A

versus:

B together with R_B.

An alternative with unchanged intrinsic properties can therefore become more or less useful when its surrounding relations change.

The derived proposition is:

When the effective contribution of a structure depends partly upon relations coordinated through it, evaluation of that structure relative to a specified continuation cannot be determined from its intrinsic properties alone.

This proposition has a boundary. If the surrounding relations make no relevant contribution to the specified continuation, then the evaluation may be determined substantially or entirely by intrinsic properties.

Freeze the bounded proposition here.

A Cross-Domain Historical Evaluation: VHS and Betamax

VHS and Betamax were incompatible consumer videocassette formats introduced into a developing home-video market.

They were not merely isolated recording technologies whose prospects could be determined by comparing technical specifications. Each required a surrounding

organization that included equipment manufacturers, distributors, retailers, prerecorded programs, repair knowledge, installed machines, and accumulated tape libraries.

The relevant comparison therefore became:

VHS together with R_{VHS}

versus:

Betamax together with R_{Beta}.

The frozen proposition is instantiated. The practical contribution of either format depended partly upon complementary relations organized around it, not solely upon the intrinsic properties of the recorder and cassette.

This does not mean that intrinsic properties were irrelevant. Recording duration, price, manufacturing capacity, licensing strategy, product range, reliability, and other differences also affected adoption. The case should not be presented as one in which an intrinsically superior Betamax inexplicably lost to an inferior VHS.

Historical scholarship instead describes an interaction among product characteristics, manufacturing and licensing strategies, complementary products, installed base, and bandwagon effects. Cusumano, Mylonadis, and Rosenbloom's study of the rivalry emphasizes strategic maneuvering and a mass-market standardization process shaped by bandwagon effects. Hiroshi Ohashi's subsequent empirical analysis found a significant role for indirect network effects in the U.S. VCR market.

The historically defensible conclusion is therefore not:

VHS prevailed despite being intrinsically inferior.

It is:

The outcome cannot be explained solely by comparing intrinsic properties of VHS and Betamax independently of the organizations and strategies constructed around them.

What the VHS/Betamax Case Adds

Once surrounding relations affect adoption, adoption itself can reconstruct those relations.

Suppose selection of A increases its installed base. That increase may encourage manufacturers, distributors, retailers, or producers of complementary goods to organize further activity through A. Those additions may then make A more useful to later participants.

The resulting relation is recursive:

adoption of A → expansion of R_A → increased contribution of A to later continuations → further adoption of A.

The condition set confronted by a later purchaser is therefore not necessarily the condition set confronted by an earlier purchaser:

$K_{t1} \neq K_{t2}$.

Earlier selections can become part of the conditions governing later selections.

This positive-feedback relation was not contained in the frozen Machiavellian proposition. It is a downstream result generated by the format case:

When adoption expands complementary relations and those relations increase the usefulness of compatibility, earlier adoption can create conditions favoring further adoption.

This relation is already recognized in economic literature as an indirect network effect or increasing return to adoption. Katz and Shapiro analyze markets in which the utility of a product depends upon the number of compatible users or products. W. Brian Arthur examines how increasing returns can make technological outcomes dependent upon historical sequences.

WMW does not independently discover network effects or path dependence. Its narrower result is that a relational proposition derived from Machiavelli leads into a mechanism already independently described and studied in another discipline.

That convergence makes the cross-domain relation less novel, but more credible.

Contraction of the Losing Organization

The recursive process can also operate through contraction.

At a given time, Betamax's technical properties need not deteriorate for its practical attractiveness to decline. If manufacturers withdraw equipment, retailers reduce inventory, prerecorded titles become less available, and users expect future support to contract, then R_{Beta} changes while the physical recording format may remain unchanged.

The case therefore produces another downstream result:

An alternative can lose practical availability because its surrounding organization contracts, even when its intrinsic properties do not deteriorate correspondingly.

Consequently:

Observed dominance does not by itself establish comprehensive intrinsic superiority.

That statement must be kept modest. It does not establish that the winning format was inferior, that the result was inefficient, or that a different history necessarily would have produced a different winner. It establishes only that dominance is a result of the complete relational process and cannot by itself identify which intrinsic property, strategic decision, or surrounding relation caused the outcome.

The case therefore exhibits historical dependence:

Previous organizational selections can become conditions governing later organizational selections.

This result belongs initially to the VHS/Betamax case and to the established literature on technological competition. It should not be attributed backward to Machiavelli without a separate textual derivation.

Inferential Path

The inferential path is:

Machiavelli's analysis of power distributed through surrounding political relations → removal of princes and political alliances → effective strength can be partly relationally produced → frozen relational proposition → VHS/Betamax → adoption can expand complementary relations → expanded relations can favor further adoption → contraction of surrounding relations can reduce the practical availability of the losing alternative.

This is not simply an analogy between Renaissance alliances and videocassette formats.

But neither is the entire feedback mechanism extracted from Machiavelli. Machiavelli supplies the relational starting proposition; the format case and existing economic theory supply the adoption dynamics.

A Comparative Stress Case: Blu-ray and HD DVD

Blu-ray and HD DVD provide a useful second case, but not a replication in the strict sense.

The case was not selected blindly, no formal replication protocol was specified, and both cases involve incompatible home-video formats. It therefore does not demonstrate survival in a radically different domain.

Its value is narrower. It asks whether the relational proposition remains useful in a later technological generation with a different configuration of manufacturers, studios, retailers, playback devices, and strategic commitments.

Again, the relevant alternatives were not merely:

Blu-ray

versus:

HD DVD.

They were:

Blu-ray together with $R_{Blu\text{-}ray}$

versus:

HD DVD together with $R_{HD\text{-}DVD}$.

The surrounding relations included hardware manufacturers, movie studios, retailers, distributors, playback devices, installed equipment, consumers, and expectations of continuing support.

The history did not consist solely of gradual consumer adoption. Large organizational decisions could reconstruct the field abruptly. Incorporation of Blu-ray drives into PlayStation 3 consoles affected the installed base. Studio commitments affected content availability. Retailer decisions affected access and expectations. Warner Bros.' decision to support Blu-ray exclusively was followed by further changes in retailer and distributor support, and Toshiba discontinued development and manufacture of HD DVD equipment in 2008.

Scholarship comparing the two format wars similarly emphasizes technological differences, strategic alliances, complementary products, software provision, and installed base rather than a single autonomous network mechanism.

The Blu-ray/HD DVD case therefore supports the frozen proposition:

The effective contribution of a format depended partly upon the organizations and complementary relations coordinated through it.

It also qualifies the VHS-derived feedback result. Surrounding organization need not grow only through numerous incremental consumer choices. A small number of strategically positioned complementors can produce a large and rapid reconstruction of the condition set.

This yields a further downstream proposition:

Where a structure depends upon complementary organizations, the decision of a highly consequential complementor can reconstruct its practical availability disproportionately.

That proposition was not frozen before the case and should not be credited to Machiavelli. It is a successor result available for later evaluation.

Analytical Significance

The weak argument would be:

Machiavelli says alliances matter; format alliances also matter.

The stronger and more disciplined result is:

1. Machiavelli supports a relational account of effective strength.
2. That relation remains evaluable after political particulars are removed.
3. VHS/Betamax instantiates the relational proposition.
4. The case then exposes recursive adoption, network effects, and historically changing conditions.
5. Blu-ray/HD DVD shows that a later format competition also depended upon surrounding relations, while revealing the disproportionate effects of strategically important complementors.

None of this shows that Machiavelli anticipated network economics.

It shows that a relation derived from his political analysis converges with an established explanation of technological competition. The contribution of WMW is the derivational route and the disciplined separation of what comes from Machiavelli, what comes from the cross-domain case, and what was already known through another analytical tradition.

Source Note

Michael A. Cusumano, Yiorgos Mylonadis, and Richard S. Rosenbloom, “Strategic Maneuvering and Mass-Market Dynamics: The Triumph of VHS over Beta,” *Business History Review* 66, no. 1 (1992): 51–94.

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CHAPTER 8 — HYPERREALISM AND EXPRESSIONISM

In Chapter V of *The Prince*, Machiavelli considers the difficulty of holding a city accustomed to liberty. Conquest does not necessarily eliminate the relations through which its former organization remains available. Its laws, memory, and established practices may continue to support restoration.

Chapter VI presents the complementary problem from the position of the founder. A new order encounters opposition from those whose positions depended upon the former order, while those who might benefit from the new order initially provide uncertain support.

The old and new organizations are therefore not consequential merely because they differ. Their possible continuations intersect through authority, allegiance, institutions, and organized support. Continuation of one can alter conditions required for continuation of the other.

The political relation is directed. An existing order may threaten a new order; a new order may threaten an existing order; or each may threaten the other differently. Symmetry is not required.

An AC Weakening

Remove the prince, conquest, territory, political rule, rebellion, military force, and the particular objectives of political actors.

Let O_1 and O_2 be distinguishable organizations. Let C_2 be a specified continuation supported through O_2 under conditions K.

The availability or continuation of O_1 threatens C_2 only if it adversely alters some relation or condition required for C_2.

Difference alone is therefore insufficient:

$O_1 \neq O_2$ does not establish Threat(O_1, O_2, C_2, K).

Nor does the mere presence of O_1 establish a threat. O_1 must alter the conditions for C_2 adversely or make their preservation less available under K.

The derived proposition is:

An alternative organization threatens a specified continuation only where its availability or continuation adversely alters relations or conditions required for that continuation.

This proposition has a boundary. If O_1 and O_2 differ but neither adversely affects conditions required by the other's continuation, the threat relation is not instantiated.

Weaker Still

Remove the assumption that the organizations are alternatives for the same position.

What remains is a directed, continuation-relative relation:

Organizational difference does not establish mutual consequence; mutual consequence does not establish adverse consequence; and adverse consequence must be evaluated relative to a specified continuation and condition set.

The distinction among these relations must be preserved:

- Difference requires only distinguishability.
- Threat requires an adverse effect upon conditions required for a specified continuation.
- Competition requires an additional relation through which distinguishable continuations become negatively coupled.

Freeze the threat proposition and its boundary here.

Do not yet assume that every threat constitutes competition or that every competitive relation must be symmetric.

A Cross-Domain Boundary Test: Hyperrealism and Expressionism

Consider Hyperrealism and Expressionism as distinguishable organizations of pictorial practice.

They differ in the relations they establish among artist, subject, pictorial surface, technique, and resulting image. The test does not treat them as contemporaneous historical rivals. Hyperrealism developed much later than historical Expressionism; the comparison deliberately concerns two distinguishable pictorial organizations rather than a documented confrontation between movements. Photorealism, closely associated with Hyperrealism, emerged in the late 1960s and early 1970s.

Let:

O_h = Hyperrealist pictorial practice

and:

O_e = Expressionist pictorial practice.

Clearly:

$O_h \neq O_e$.

But the frozen proposition prohibits deriving threat from difference alone.

Suppose:

C_h = continuation of Hyperrealist pictorial practice

and:

C_e = continuation of Expressionist pictorial practice.

Production of a Hyperrealist painting does not, merely by occurring, adversely alter the conditions required for continued Expressionist painting. The converse also does not follow.

Accordingly:

$O_h \neq O_e$

does not establish:

Threat(O_h, O_e, C_e, K)

or:

Threat(O_e, O_h, C_h, K).

The case does not instantiate the positive threat relation. It instead satisfies the frozen boundary: difference is present while adverse mutual consequence is absent.

That is not a failed test. It is a boundary result.

Reconstruction Under an Expanded Condition Set

Now place the practices within an art-world organization that allocates exhibition space, acquisitions, funding, instruction, critical attention, or institutional prestige.

The containing organization alone does not establish competition. Nor does scarcity in the abstract. A negatively coupled relation must actually be present.

Suppose a gallery has one exhibition period available, and allocation of that period to Hyperrealist work makes it unavailable to Expressionist work. Continuation of one

practice through this particular institutional relation then adversely affects conditions available to the other.

Under the expanded conditions K′:

K′ = K together with a limiting allocation relation.

The threat relation may now be instantiated:

Threat(O_h, O_e, C_e, K′).

If each practice is being continued through pursuit of the same limiting allocation, a competitive relation may also be present.

This competition is not intrinsic to Hyperrealism and Expressionism. It is produced through their participation in a containing organization whose allocation structure negatively couples their continuations.

The ecological conception of competition provides a useful independent comparison: organisms need not intentionally oppose one another; competition can arise because their use of a limited shared resource adversely affects one another's continuation.

Result of the Boundary Test

The test produces:

1. non-instantiation under K;
2. contextual instantiation under K′.

Under K, the practices differ without threatening one another.

Under K′, an additional allocation relation can make their continuations adversely consequential to one another.

The second result does not reverse the first. It identifies the organizational addition that changes the relation:

K → K′

and thereby makes threat or competition available.

The correct conclusion is therefore:

Alternative organizations need not be threatening or competitive organizations.

And:

A competitive relation can be produced by negatively coupled relations within a containing organization rather than by the intrinsic properties of the organizations described as competitors.

Continuation-Indexed Competition

Competition cannot be adequately represented as an intrinsic binary relation:

$\mathrm{Comp}(O_1, O_2)$.

Nor is $\mathrm{Comp}(O_1, O_2, C, K)$ sufficient where the organizations support distinguishable continuations. A single C conceals whose continuation is affected.

Use:

$\mathrm{Comp}(O_1, O_2, C_1, C_2, K)$.

This indicates that O_1 and O_2 support distinguishable continuations C_1 and C_2 whose conditions are negatively coupled under K.

The relation may be asymmetric. O_1 may substantially reduce availability of C_2 while O_2 produces only a minor effect upon C_1. If that direction matters, it should be stated explicitly rather than hidden inside a symmetric predicate.

Accordingly:

$\neg\mathrm{Comp}(O_h, O_e, C_h, C_e, K)$

may hold while:

$\mathrm{Comp}(O_h, O_e, C_h, C_e, K')$.

This is not a change in the intrinsic difference between O_h and O_e. It is a change in the organization through which their continuations are evaluated.

Difference, Incompatibility, Threat, and Competition

The case reveals four distinguishable relations.

Difference

O_1 and O_2 are distinguishable:

$O_1 \neq O_2$.

Incompatibility

O_1 and O_2 cannot both be preserved relative to some specified continuation and conditions.

Threat

Availability or continuation of O_1 adversely affects conditions required for C_2.

Competition

O_1 and O_2 support distinguishable continuations whose conditions are negatively coupled through some shared or containing relation.

These relations must not be collapsed.

Difference does not establish incompatibility, threat, or competition.

Incompatibility does not by itself establish a competitive process. Two structures may be logically or technically incapable of joint instantiation without either organization competing for anything.

Threat does not necessarily establish competition. A structure can threaten another continuation without pursuing or depending upon a competing continuation of its own.

Competition does not require complete incompatibility. Competing organizations may continue simultaneously while adversely affecting one another's access to some limiting relation.

Inferential Path

The disciplined derivation is:

Machiavelli's analysis of old and new orders → continuation of one order can adversely alter conditions required by another → removal of political particulars → frozen, directed threat relation → Hyperrealism and Expressionism → difference without threat → addition of a negatively coupled allocation relation → contextual instantiation of threat and competition.

The art case therefore does more than provide an unusual analogy. It prevents the political source from silently contributing a condition that disappears during weakening.

Political alternatives frequently contest authority, territory, allegiance, or control of institutions. Their mutually consequential relation can therefore appear automatic. Once those political particulars are removed, however, alternative organization no longer entails competition.

Verdict

The boundary test produces non-instantiation followed by contextual instantiation.

Hyperrealism and Expressionism are distinguishable organizations of pictorial practice, but that difference alone does not establish threat or competition:

$O_h \neq O_e$ does not establish Comp(O_h, O_e, C_h, C_e, K).

A competitive relation becomes available only under additional conditions that negatively couple their continuations—for example, an institutional allocation in which increased availability to one reduces availability to the other.

The initial non-instantiation is not repaired by redescribing the practices. It remains valid under K. The later instantiation occurs under K′ and therefore constitutes a different organizational result.

This establishes an important stopping condition for the WMW method:

AC Weakening does not guarantee cross-domain instantiation. A frozen proposition may remain evaluable in another domain while the selected organizations fail to instantiate it.

The method may therefore return *not instantiated* without changing the frozen proposition or adding undeclared conditions merely to produce a successful result.

CHAPTER 9 — EVOLUTION

In Chapter XXV of *The Prince*, Machiavelli considers why a mode of action that produces success under one set of conditions can produce failure when circumstances change.

His analysis turns upon a mismatch between a retained mode and altered conditions. One person proceeds cautiously and another impetuously; one relies upon patience and another upon force. Success occurs when the mode of proceeding corresponds sufficiently to the conditions in which it operates.

Success itself, however, does not guarantee continued correspondence.

A person may prosper while conditions favor the mode through which that prosperity was achieved. If the conditions change while the mode remains, the same conduct may cease to produce the same result. Machiavelli emphasizes the difficulty people have in departing from dispositions and practices through which they previously succeeded.

The relation is therefore more precise than the general instruction to adapt to circumstances.

Under conditions K_1:

M contributes to C.

Conditions then change:

$K_1 \rightarrow K_2$.

The previous contribution of M under K_1 does not by itself establish:

M contributes to C under K_2.

But the mere fact that K_1 and K_2 differ is insufficient. The changed conditions must include something relevant to the relation through which M contributed to C. If the difference between K_1 and K_2 is irrelevant to that relation, the earlier result may remain transferable.

Machiavelli expresses the problem politically and psychologically. The underlying dependency can nevertheless be evaluated after those particulars are removed.

An AC Weakening

Remove the prince, *fortuna*, political success, personality, prudence, impetuosity, and deliberate action.

Let O be an organization whose specified continuation is C. Let M be a configuration of O that contributes to C under conditions K_1.

The contribution of M is condition-indexed:

$E(M, C, K_1)$.

If relevant conditions change from K_1 to K_2, the previous effectiveness of M does not establish its effectiveness under the new conditions:

$E(M, C, K_1)$ does not entail $E(M, C, K_2)$.

This does not establish that M will become ineffective under K_2. It establishes only that effectiveness under K_1 cannot be transferred to K_2 without evaluating the changed conditions relevant to the M–C relation.

Historical contribution also does not establish present contribution:

M contributed to C under K_1 does not entail that M contributes to C under K_2.

The introduction of necessity is unnecessary here. Establishing that M contributed to C under K_1 would not establish that M was necessary for C even under K_1. Other configurations might also have preserved C.

A configuration can therefore remain instantiated after the conditions that made it effective have changed. If another configuration M′ would contribute more effectively to C under K_2, continued retention of M may reduce the availability of C even though M remains associated with previous success.

The weakened proposition is:

A configuration's contribution to a specified continuation is relative to the conditions through which that contribution is produced.

And:

Previous contribution under one condition set does not establish continued contribution after causally relevant conditions change.

Machiavelli's problem can therefore be separated from the human difficulty of deliberately changing one's behavior.

The mismatch itself does not require intention.

Weaker Still

Remove the requirement that O deliberately selects, retains, or changes M.

M need only be instantiated under conditions in which it has consequences for C. Neither M nor O must recognize those consequences.

If relevant conditions change, the consequences of the retained configuration may also change without any component intentionally responding to the change.

The resulting proposition is:

Causally relevant changes in conditions can change the continuational consequence of an unchanged configuration.

This proposition has two important boundaries.

First, a change from K_1 to K_2 does not establish a change in the consequence of M unless the altered conditions are relevant to the M–C relation.

Second, the proposition does not predict the direction of the change. M may become more advantageous, less advantageous, neutral, or remain unaffected.

Freeze the bounded proposition here.

A Cross-Domain Agency-Removal Test: Evolution by Natural Selection

Consider a population containing heritable variation.

Organisms differ in their configurations of traits. Under specified environmental conditions, bearers of some heritable variants may survive or reproduce differently from bearers of others.

Let M represent a heritable variant or sufficiently stable heritable configuration. Under K_1, suppose organisms bearing M have greater expected reproductive success than relevant alternatives.

The relation is not:

M reproduces.

It is:

bearers of M reproduce differentially under K_1.

Where that difference is heritable, M may consequently become more frequent in later generations. Variation, differential reproduction, and heredity are the minimal elements ordinarily used to explain evolution by natural selection.

No organism needs to know that M is advantageous.

No organism needs to choose M.

The population requires no governing objective and no intention to reconstruct itself.

Now let relevant environmental conditions change:

$K_1 \rightarrow K_2$.

The frozen proposition predicts only that the previous consequence of M cannot automatically be transferred to K_2.

Under K_1, bearers of M may have greater expected reproductive success:

M under $K_1 \rightarrow \Delta$ reproductive contribution > 0.

Under K_2, the same variant may be neutral or disadvantageous:

M under $K_2 \rightarrow \Delta$ reproductive contribution $= 0$

or:

M under $K_2 \rightarrow \Delta$ reproductive contribution < 0.

The variant need not have changed. Its relation to the conditions has changed.

This instantiates the frozen proposition:

Causally relevant changes in conditions can change the continuational consequence of an unchanged configuration.

The result survives after deliberate action, political judgment, foresight, personality, and strategy have been removed.

What Natural Selection Adds

Evolutionary populations contain variation among heritable configurations. Changed conditions therefore need not alter only the consequences of a single retained configuration. They may alter the relative reproductive consequences of several configurations.

Suppose M_1, M_2, and M_3 occur in a population.

Under K_1, bearers of M_1 have the greatest expected reproductive contribution.

Under K_2, bearers of M_3 have the greatest expected reproductive contribution.

If these differences are heritable and sufficiently persistent, differential reproduction can contribute to a change in the population frequencies of the variants:

$K_2 \rightarrow$ differential reproductive success $\rightarrow \Delta p(M_1), \Delta p(M_2), \Delta p(M_3)$.

Here $p(M_i)$ designates the frequency of M_i in the population.

This relation must remain probabilistic and bounded. Natural selection is not the only process affecting population composition. Mutation, recombination, genetic drift, migration, mating structure, and developmental effects may also alter the distribution of variants. Selection can contribute to reconstruction of the population without uniquely determining its later organization.

The cross-domain case therefore adds:

Conditions can contribute to organizational reconstruction by differentially affecting the continuation and transmission of available configurations.

This result should not be attributed backward to Machiavelli.

Machiavelli's actors alter their conduct—or fail to do so. Natural selection requires no corresponding act of self-reconstruction. Population-level change can emerge because bearers of heritable variants reproduce differently under specified conditions.

Distinguishable Continuations

The test becomes especially severe when the relevant continuation is specified.

Chapter XXV concerns an actor whose political continuation is threatened and whose conduct might, at least conceptually, be altered to preserve it.

Natural selection contains no necessary equivalent of that unified actor and objective.

At least three continuations must be distinguished:

C_i = continued existence or reproduction of an individual bearer;

C_m = continued transmission or increased frequency of heritable variant M;

C_p = persistence of the population.

These continuations are related, but results do not automatically transfer among them.

A trait may reduce an individual's longevity while increasing its reproductive contribution. A variant may increase in frequency while contributing adversely to some aspect of population persistence. Selection among cell lineages can even favor cellular proliferation that damages the containing organism, as occurs in cancer. This is one reason levels of selection must be kept distinct.

Accordingly:

$E(M, C_i, K)$ does not entail $E(M, C_m, K)$.

$E(M, C_m, K)$ does not entail $E(M, C_p, K)$.

Nor do the reverse entailments follow without additional relations.

The evolutionary test therefore forces greater precision into the weakened proposition:

A continuational consequence established at one organizational level does not thereby establish the same consequence at another level.

This does not settle disputes about levels or units of selection. It explains why the relevant level cannot be left unspecified.

Results Added by Evolution

The cross-domain test produces three downstream results.

First:

Changed conditions can contribute to reconstruction through differential continuation rather than deliberate modification.

Second:

A configuration's reproductive consequence for its bearers must be distinguished from the resulting change in the population distribution of that configuration.

Third:

A result established relative to continuation at one organizational level does not automatically transfer to continuation at another level.

These results are consistent with established evolutionary theory rather than discoveries about evolution produced by WMW. Their importance here is methodological: the evolutionary case exposes distinctions that remained partly hidden in the political source.

Inferential Path

The complete path is:

Chapter XXV → success of a mode depends upon correspondence with conditions → relevant conditions can change while the mode remains → removal of prince, personality, strategy, and deliberate adaptation → the continuational consequence of a configuration is condition-relative → frozen bounded proposition → heritable variation and differential reproduction → changed conditions alter the reproductive consequences of retained variants → differential reproduction can contribute to population-level reconstruction without deliberate modification → results must remain indexed to distinguishable organizational levels.

Verdict

The frozen proposition survives the agency-removal test.

A heritable variant that contributed to differential reproductive success under K_1 need not make the same contribution under K_2 when causally relevant environmental conditions have changed. Neither the organism nor the population must recognize the change or deliberately reconstruct itself.

Evolution then adds a mechanism absent from the Machiavellian derivation:

Changed conditions can contribute to reconstruction of a population by differentially affecting the reproduction and transmission of available heritable variants.

It also exposes a boundary:

Reproductive continuation of an individual, transmission of a variant, and persistence of a population are distinguishable continuations. A result established for one does not automatically establish a result for another.

Evolution does not show that Machiavelli anticipated natural selection. It shows that a bounded condition-relative continuation relation extracted from his political analysis remains evaluable after intentional adaptation has been removed entirely.

CHAPTER 10 — ECOLOGICAL SYSTEMS

In Chapter III of *The Prince*, Machiavelli does not evaluate political actors solely according to their individual strength. He examines the relations among stronger and weaker powers and the consequences of altering those relations.

A new ruler entering a territory should protect the weaker powers without increasing their strength, restrain the stronger powers, and prevent another great power from entering. Machiavelli's concern is the political organization produced through their relative positions. A lesser power that appears insignificant when considered independently may matter because of the relations it occupies within the political field.

Louis XII's errors demonstrate the inverse. By weakening some powers, strengthening others, and admitting Spain into Italy, Louis did not merely alter several independent actors. He reconstructed the relational environment within which French power itself had to continue.

The consequence of removing or weakening a participant therefore cannot necessarily be inferred from the apparent importance of that participant considered in isolation.

An organization can matter because other continuations depend upon relations in which it participates.

The immediate political proposition is:

Changing the availability of one participant can alter conditions governing the continuation of other participants when their continuations are relationally connected.

An AC Weakening

Remove France, Italy, princes, alliances, conquest, strategy, and political power.

Let O contain distinguishable components A_1, A_2, … A_n connected through a set of relations R under conditions K.

The availability of each component's continuation does not by itself establish continuation of the containing organization:

availability of C_1, C_2, … C_n does not establish C_o.

Conversely, loss or weakening of one component does not by itself establish failure of C_o. The result depends upon the relations required for the specified continuation.

Suppose modification X acts directly upon A_1:

$X \rightarrow \Delta A_1$.

Suppose continuation C_2 depends upon a relation R_{12} involving A_1 and A_2. A change in A_1 may then alter the availability of C_2 without X acting directly upon A_2:

$X \rightarrow \Delta A_1 \rightarrow \Delta R_{12} \rightarrow \Delta$ availability of C_2.

If C_3 subsequently depends upon a relation involving A_2, the consequence may propagate further:

$\Delta A_1 \rightarrow \Delta R_{12} \rightarrow \Delta A_2 \rightarrow \Delta R_{23} \rightarrow \Delta$ availability of C_3.

The arrows here represent a possible causal propagation through specified relations, not logical implication.

The intervention is local in a relational sense: X acts directly upon A_1. Its consequences are nonlocal when they reach components or continuations upon which X did not act directly.

This yields:

The organizational consequence of altering a component depends partly upon the relations through which specified continuations depend upon that component.

And:

A modification directed at one component can alter other continuations through propagation across dependency relations.

Neither result guarantees propagation. The relevant relation may be redundant, weak, compensated, or unnecessary for the continuation being evaluated. The direction and magnitude of the downstream consequence must also be established rather than inferred from connectivity alone.

Weaker Still

The preceding formulation assumes that the significance of A_1 can be represented through identifiable pairwise dependencies.

Remove that requirement.

A component may participate in multiple relations whose combined consequence is not represented by any single pairwise connection. Altering it may simultaneously change resource availability, protection, accessibility, competition, timing, habitat, or other conditions.

The significance of a component is therefore not an intrinsic property of that component. It is indexed to:

- the continuation being evaluated;
- the relations in which the component participates;
- the applicable conditions;
- the evaluation horizon;
- the availability of redundant or compensating relations.

The broader proposition is:

The significance of a component to a specified organizational continuation cannot be determined solely from properties of the component considered independently of its relations.

The formulation:

A local change can reconstruct conditions beyond the location at which it occurs

is weaker, but insufficiently bounded. Almost any causal system could satisfy it.

Freeze the more informative pair:

The consequence of altering a component relative to a specified continuation depends partly upon the relations through which that continuation depends upon the component.

A modification directed at one component can produce consequences elsewhere in a relational organization by altering relations required for other continuations.

A Cross-Domain Ecological Evaluation: Sea Otters and Kelp Forests

Consider a coastal ecological system containing sea otters, sea urchins, kelp, and organisms whose continuations depend directly or indirectly upon kelp-forest habitat.

Under documented conditions in parts of the North Pacific, sea otters prey upon sea urchins, while sea urchins consume kelp. Otter predation can affect both urchin abundance and grazing behavior.

In simplified form:

greater otter predation → reduced urchin grazing pressure → greater kelp availability.

A sufficiently large decline in otter abundance can therefore have consequences that do not remain confined to the otter population:

reduced otter predation → increased urchin grazing pressure → reduced kelp.

Reduced kelp can then alter habitat structure, food availability, shelter, productivity, and other conditions affecting organisms that were not directly acted upon by whatever produced the otter decline.

This is a well-established trophic-cascade pattern, especially in Alaska and the Aleutian Islands. But it is not a universal law governing every kelp system. The magnitude and timing of otter effects vary with habitat, alternative predators, urchin behavior, productivity, and other local conditions. Long-term research on sea-otter reintroduction has found explicitly dynamic and context-dependent effects. Estes and Duggins' foundational study; 2025 PNAS study.

The case should therefore be stated conditionally:

Under conditions in which otter predation substantially regulates urchin grazing and kelp continuation depends upon that regulation, a decline in otters can propagate through urchins to kelp and to kelp-dependent relations.

This conditional form is stronger methodologically than presenting the familiar three-level cascade as universal.

Evaluation of the Frozen Propositions

The first frozen proposition is instantiated.

The organizational consequence of changing otter availability cannot be determined from the isolated properties or numerical abundance of otters. It depends upon relations connecting otter predation, urchin activity, kelp abundance, and kelp-dependent continuations.

The second frozen proposition is also instantiated.

Whatever causes an otter decline need not act directly upon urchins, kelp, or kelp-dependent organisms. Its consequences can reach them through altered ecological relations.

This is relationally nonlocal propagation.

But the case also illustrates the boundary. Where other predators regulate urchins, where urchin grazing is not limiting kelp, or where other conditions dominate kelp continuation, an otter decline may not produce the predicted cascade. Empirical studies have reported locations in which reduced otter density did not produce a corresponding change in kelp abundance. USGS study.

The case therefore supports the bounded proposition, not an unconditional rule that otter decline entails kelp decline.

Endogenous Reconstruction of Conditions

Kelp is not merely another population in the sequence. Kelp forests help constitute habitat structure and other conditions encountered by participating organisms.

A propagated change can therefore alter the conditions governing later relations within the same ecological system.

Let O_t designate the ecological organization at one indexed state. Changes propagated through O_t may contribute to a later condition set K_{t+1}:

$\Delta O_t \rightarrow \Delta K_{t+1}$.

Those reconstructed conditions may then alter relations within the subsequent organization:

$\Delta K_{t+1} \rightarrow$ altered consequences within O_{t+1}.

This yields a recursive relation:

$O_t \rightarrow K_{t+1} \rightarrow O_{t+1}$.

The condition set is therefore not necessarily imposed entirely from outside the organization. Some later conditions may be endogenously produced through earlier relations within it.

The ecological evaluation adds:

Propagation through a relational organization can reconstruct conditions governing later propagation through that organization.

This does not mean that every component should be counted simultaneously as both organization and condition. The indexing distinguishes the roles: a component or relation within O_t contributes to a condition encountered by O_{t+1}.

Distinguishable Ecological Continuations

The ecosystem should not be treated as an actor possessing a single objective.

An evaluator may specify:

C_o = persistence of the sea-otter population;

C_u = persistence of the sea-urchin population;

C_k = persistence of kelp-forest organization;

C_p = persistence of a specified community composition;

C_b = maintenance of specified biodiversity;

C_e = continuation of some coastal ecological organization, regardless of its particular regime.

These continuations are distinguishable. Satisfying one does not automatically satisfy the others.

Increased urchin abundance may contribute to some specified continuation of the urchin population while degrading conditions required for C_k. Kelp expansion may support some populations while altering conditions available to others.

Accordingly:

preservation of C_u does not establish preservation of C_k;

preservation of C_k does not establish preservation of C_o;

preservation of C_e does not establish preservation of C_k.

This last distinction is particularly important.

If an otter decline contributes to a transition from a kelp forest to an urchin barren, a coastal ecological system may continue to exist while the specified kelp-forest

organization does not. The result is organizational reconstruction, not necessarily disappearance of every containing organization.

Consequently, the statement:

The ecosystem benefits

is inadmissibly underspecified.

It may mean preservation of kelp, biodiversity, productivity, a particular community configuration, a valued human use, or merely continued ecological activity. These are not interchangeable results.

Continuation Without Objective

The absence of a common objective does not prevent an evaluator from specifying a system-level continuation.

This distinction must be preserved:

- possessing an objective is a capability of the organization;
- specifying a continuation is an operation performed within the evaluation.

The ecosystem need not seek C_k, C_b, or C_e. Otters, urchins, and kelp need not share any objective. Nevertheless, their relations can be evaluated relative to any clearly specified continuation.

The ecological case therefore supports:

Relational organization does not require a common objective among participating components.

And:

Specification of a containing continuation does not establish that the containing organization possesses that continuation as an objective.

This is the deeper result. Distinguishable continuations can be specified for the relevant populations and organizational levels without attributing those specifications to the organisms themselves.

What Ecology Adds

The ecological evaluation produces three downstream results.

First:

A propagated change can reconstruct conditions governing later propagation through the same relational organization.

Second:

Relational organization does not require a common objective among its components.

Third:

Persistence of a containing system does not establish persistence of a specified organization within that system.

These results should not be attributed backward to Machiavelli, nor are they discoveries about ecology. Trophic cascades, ecological feedback, context dependence, and alternative community regimes are already established subjects of ecological research.

WMW's contribution is the derivational route and the separation of:

- what is extracted from Machiavelli;
- what survives weakening;
- what the ecological case instantiates;
- what the ecological case adds;
- what must remain indexed to a specified continuation.

Inferential Path

The complete path is:

Chapter III → political significance depends partly upon relations among stronger and weaker powers → altering one participant can change conditions encountered by others → removal of political actors, alliances, strategy, and intentional opposition → component significance is relation- and continuation-dependent → a directly targeted change can propagate to untargeted components → frozen propositions → sea otters, urchins, and kelp → conditional trophic propagation → reconstruction of later environmental conditions → distinguishable ecological continuations → relational organization without common objective.

Verdict

The sea-otter case instantiates the frozen propositions under specified ecological conditions.

A change directed at one component can propagate through predation, grazing, habitat formation, and other relations to affect continuations not directly acted upon. The consequence of changing the component therefore cannot be evaluated solely from its isolated properties.

The test also establishes a boundary: connectivity does not guarantee propagation, and the familiar otter–urchin–kelp cascade is context-dependent.

Ecology then adds the chapter's most important results:

A relational organization can reconstruct conditions governing its later operation.

Relational organization does not require a common objective.

A continuation specified for a containing organization need not be possessed as an objective by that organization.

The defensible conclusion is not that Machiavelli anticipated ecology or trophic cascades. It is that a relational consequence extracted from his political analysis remains evaluable after politics, strategy, authority, intentional coordination, and shared objective have been removed.

PART IV — WHAT SURVIVED?

CHAPTER 1 — CROSS-DOMAIN SURVIVAL

The evaluations in Part III do not establish that Renaissance politics, cancer treatment, software development, measurement standards, technological formats, artistic practices, evolution, and ecology are instances of one phenomenon.

They establish something narrower.

The proportion of positive instantiations among these cases has no confirmatory significance. The cases were selected for analytical value and with prior awareness of their possible relevance; their outcomes therefore cannot estimate how frequently the frozen relations obtain. Each evaluation establishes only whether the specified relation instantiates under that case's stated conditions.

Relations derived from Machiavelli's political analyses remained evaluable after the political particulars through which those relations were expressed had been progressively removed.

Survival does not require positive instantiation in every case. A proposition survives cross-domain evaluation when it can be evaluated without being altered merely to fit the new domain. The result may be:

- positive instantiation;
- non-instantiation;
- instantiation only under additional conditions;
- exposure of an omitted index or boundary.

The surviving relations differ from chapter to chapter.

From Savonarola survives the distinction between initial acquisition of a continuation and retained availability of the relation required to preserve it.

From the election case survives the distinction between continuation of competing components and continuation of the organization containing their competition.

From Louis XII survives the proposition that an intervention selected relative to one continuation can alter conditions affecting other continuations.

From the thyroid-cancer evaluation survives the distinction among localized effect, containing continuation, and the conditions bounding an intervention's admissibility.

From CCIBASIC survives the distinction between availability of a required capability and continued dependence upon the particular relation through which that capability was previously supplied.

From the inch-based and metric standards evaluation survives the proposition that replacement difficulty can depend upon required relations already organized through an instantiated structure.

From the VHS/Betamax evaluation survives the narrower proposition that the effective contribution of a structure can depend partly upon relations organized around it, rather than solely upon its intrinsic properties.

The recursive relation:

adoption → expanded surrounding organization → increased contribution to later continuations → further adoption

was not frozen before the VHS/Betamax case. It is a downstream result produced by the cross-domain evaluation and independently recognized in the literature on network effects and increasing returns.

Hyperrealism and Expressionism produce a different form of survival. Their difference does not instantiate the frozen threat relation under the initial condition set K. Threat and competition become available only under an expanded condition set K′ containing a negatively coupled allocation relation.

The initial non-instantiation preserves rather than defeats the frozen boundary:

$O_h \neq O_e$ does not establish Threat(O_h, O_e, C_e, K).

Evolution preserves the bounded proposition that causally relevant changes in conditions can change the continuational consequence of an unchanged configuration. The proposition remains evaluable after deliberate action and intentional adaptation are removed.

Ecology preserves the proposition that alteration of one component can affect other continuations through dependency relations, even where no component intentionally coordinates the process.

The ecological evaluation subsequently adds that relational organization does not require a common objective and that specification of a containing continuation does not attribute that continuation to the system as an objective.

These downstream results must remain distinguishable from the relations frozen before evaluation.

The cross-domain result is therefore not:

Machiavelli's political rules apply everywhere.

It is:

Some relations derived from Machiavelli's political analyses remain evaluable after the political actors, objects, objectives, and eventually even intentional agency through which he observed them have been removed.

Cross-domain survival is evidence of retained relational structure, not evidence of identity among domains.

CHAPTER 2 — FAILURE AND BOUNDARY

Successful instantiation alone would provide weak evidence for the method if every evaluation were permitted to succeed.

Part III therefore includes cases in which a frozen relation does not initially instantiate, instantiates only under additional conditions, or becomes coherent only after an omitted index is restored.

Non-Instantiation

The Hyperrealism/Expressionism evaluation begins with two distinguishable organizations of pictorial practice:

$O_h \neq O_e$.

Their difference alone does not establish that continuation of either adversely changes conditions required for continuation of the other.

Accordingly:

$O_h \neq O_e$ does not establish Threat(O_h, O_e, C_e, K)

or:

Threat(O_e, O_h, C_h, K).

The frozen threat relation is not instantiated under K.

The evaluation does not thereby fail. The result is *not instantiated*.

Contextual Instantiation

Threat or competition becomes available only after an additional relation is specified—for example, participation in an institutional allocation through which exhibition space, acquisition, funding, or another limiting relation becomes negatively coupled.

This produces an expanded condition set:

$K \rightarrow K'$.

Under K′, continuation of one practice may adversely alter conditions available to the other.

The later instantiation does not erase the earlier non-instantiation. The two results concern different condition sets:

$\neg Threat(O_h, O_e, C_e, K)$

may coexist with:

$Threat(O_h, O_e, C_e, K')$.

The difference between K and K′ identifies the relation missing from the initial case.

Specification Failure

Evolution produces another kind of boundary. The condition-relative proposition survives, but an apparent evolutionary result becomes equivocal if the continuation is not specified correctly.

At least three continuations must be distinguished:

C_i = continuation or reproduction of an individual bearer;

C_m = transmission or increased frequency of heritable variant M;

C_p = persistence of the population.

A result established relative to one does not automatically transfer to the others:

$E(M, C_i, K)$ does not entail $E(M, C_m, K)$.

$E(M, C_m, K)$ does not entail $E(M, C_p, K)$.

This is not non-instantiation of the condition-relative proposition. It is failure to specify the result at the organizational level where it occurs.

Attribution Failure

Ecology produces a related boundary.

An evaluator may specify a containing continuation such as persistence of kelp-forest organization, biodiversity, or a particular community configuration. That does not establish that the ecological system possesses the specified continuation as an objective.

Specification of C_o does not establish possession of C_o as an objective by O.

Relational organization can remain evaluable after common objective is removed.

Forms of Boundary Output

Part III therefore exposes at least four distinguishable outputs:

Non-instantiation: The relation required by the frozen proposition is absent under the specified conditions.

Contextual instantiation: The relation appears only after an additional condition is explicitly introduced.

Specification failure: An apparent result becomes equivocal or disappears when the continuation, conditions, horizon, or organizational level is specified correctly.

Attribution failure: A continuation introduced by the evaluator is incorrectly attributed to the evaluated organization as its own objective.

These are not defects in the method. They are among its required outputs.

A method that can return only *instantiated* cannot reliably distinguish discovery from accommodation.

CHAPTER 3 — SPECIFICATION ROBUSTNESS

The most persistent result across Part III may not be any individual proposition derived from Machiavelli.

It may be the repeated survival of specification itself.

Again and again, apparently categorical statements become unstable when their continuations and conditions remain implicit.

Necessary becomes:

N(X, C, K).

Effective becomes:

E(X, C, K).

Admissible becomes:

A(X, C, K).

A directed threat relation requires identification of the threatened continuation:

Threat(O_1, O_2, C_2, K).

Competition among distinguishable continuations requires at least:

Comp(O_1, O_2, C_1, C_2, K).

The additional continuation index is necessary because competition may concern effects upon different continuations supported through O_1 and O_2. A single C can conceal whose continuation is being altered.

Independence in CCIBASIC also becomes continuation-relative. Reduced dependency relative to the production-testing continuation C_t does not establish reduced dependency relative to the maintenance continuation C_i.

The same discipline applies temporally. A result established under K_1 does not automatically transfer to K_2:

R(X, C, K_1) does not entail R(X, C, K_2).

Here R is a schematic placeholder for the relation being evaluated, not a newly defined AC operator.

The discipline also applies across organizational levels:

R(X, C_1, K) does not entail R(X, C_2, K).

This is not merely a formal preference for additional subscripts.

The cross-domain evaluations repeatedly show what becomes concealed when the indices are omitted.

The shepherd's continuation can be confused with the flock's.

Campaign victory can be confused with continuation of the electoral system.

Successful cancer treatment at t_1 can be confused with the patient's later condition set at t_2.

Availability of CCIBASIC for producing test programs can be confused with availability of the expertise required to maintain CCIBASIC itself.

Adoption of metric standards for new designs can be confused with admissibility of removing inch-based standards from legacy support.

Individual reproductive contribution can be confused with transmission of a heritable variant or persistence of a population.

Persistence of some coastal ecological organization can be confused with persistence of kelp-forest organization.

These distinctions do not establish that every organizational predicate must always have exactly the form R(X, C, K). Some relations may require additional indices, fewer indices, directed arguments, a horizon, or a specification more complex than a single continuation.

The supported conclusion is narrower:

Where changing the continuation, conditions, horizon, direction, or organizational level changes the result, suppressing that index removes analytically relevant information.

The robustness result is therefore:

Relations that appear categorical may become evaluable without equivocation only after the continuation and conditions relative to which they hold are specified.

The requirement repeatedly survives movement across otherwise dissimilar domains.

That is the specification discipline preserved by Part III.

CHAPTER 4 — CANDIDATE RELATIONAL FORMS

"Relational form" is deliberately weaker than "organizational invariant." The evaluations do not establish universal laws, nor does appearance in several domains by itself establish invariance.

A candidate relational form, as used here, is a relation that remains evaluable after substantial removal or replacement of the domain-specific particulars through which it was first identified. Its survival does not establish truth across all organizations, invariance under a defined class of transformations, or universality.

The cases support candidacy for further evaluation, not invariance or universality.

Several candidates emerge.

Continuation-Relative Evaluation

Effectiveness, necessity, admissibility, independence, threat, competition, and preservation cannot automatically be transferred from one continuation to another.

In schematic form:

$R(X, C_1, K)$ does not entail $R(X, C_2, K)$.

This formulation does not claim that transfer never occurs. It claims that transfer requires establishment of the relevant relation between C_1 and C_2.

Condition-Relative Consequence

A configuration or intervention that contributes to a continuation under K_1 need not make the same contribution under K_2 when causally relevant conditions have changed:

$R(X, C, K_1)$ does not entail $R(X, C, K_2)$.

Again, $K_1 \neq K_2$ alone is insufficient. The changed conditions must be relevant to the relation being evaluated.

Relational Availability

A continuation may depend not upon possession of a component or capability considered independently, but upon continued availability of relations through which its required consequences are supplied.

Savonarola, auxiliary arms, CCIBASIC, instantiated standards, technological formats, and ecological dependencies exhibit different forms of this relation.

This does not make those cases instances of one phenomenon. It identifies a relational dependency preserved across their differences.

Propagation Beyond the Targeted Component

A modification directed at one component can alter continuations associated with components upon which it did not act directly.

The compact proposition exposed through Louis XII is:

An intervention selected for one continuation can produce consequences for another continuation when it alters conditions accessible to both.

Cancer and ecology add that propagation can reconstruct later conditions under which the containing organization operates.

Propagation is not guaranteed. It requires relevant relations and may be interrupted by redundancy, compensation, weak coupling, or other conditions.

Historical and Endogenous Reconstruction

Continuation occurs within conditions that may be partly produced by previous continuation:

K_{t1} need not equal K_{t2}.

The organization evaluated later may differ not because the focal component changed intrinsically, but because previous actions, adoptions, interventions, or differential continuations reconstructed the surrounding relations.

Ecology adds the more explicitly recursive form:

$O_t \rightarrow K_{t+1} \rightarrow O_{t+1}$.

Some conditions governing later organization can therefore be generated through earlier organization rather than imposed wholly from outside it.

Local and Containing Divergence

A result favorable to continuation of a component need not contribute favorably to continuation of a containing organization:

contribution to C_{local} does not entail contribution to $C_{containing}$.

The two continuations may be compatible, independent, or opposed. Divergence must be established rather than presumed.

The election case makes this distinction explicit. Cancer, evolution, and ecology prevent it from being treated as exclusively political.

Relational Consequence Over Intrinsic Property

The organizational consequence of a component cannot always be inferred from properties of the component considered independently of its relations.

This appears in Louis XII, instantiated standards, format adoption, and the otter–urchin–kelp system in substantially different forms.

The candidate proposition is not that intrinsic properties are irrelevant. It is that they may be insufficient for evaluating organizational consequence.

Organization Without Common Objective

The ecological evaluation supplies the most severe candidate:

Relational organization does not require a common objective among its components.

This must be distinguished from the stronger and unsupported claim that a containing continuation cannot be specified.

An evaluator may specify C_o without attributing C_o to O as an objective:

specification of C_o does not establish possession of C_o by O as an objective.

The distinction permits evaluation of non-agentive organizations without importing a governing purpose into them.

Status of the Candidates

These are candidate relational forms exposed through the derivation-and-evaluation procedure.

Their candidacy rests neither upon Machiavelli's authority nor upon superficial resemblance among the tested domains. It rests upon the continued evaluability of their relational forms after substantial removal of the particulars through which they were initially observed.

Further testing may:

- preserve a candidate without modification;
- restrict its boundary;
- expose an omitted index;
- divide one candidate into several relations;
- show that apparent survival resulted from an overly permissive reconstruction;
- produce non-instantiation in a domain where instantiation had been expected.

Part IV has therefore answered the first question:

What survived the controlled removal of Machiavelli's political particulars?

The answer is not a single universal proposition. It is a set of candidate relational forms, their provenance, and the boundaries exposed through cross-domain evaluation.

Part V asks the interpretive question that follows:

What does that survival establish about the character and continuing force of Machiavelli's political insight?

PART V — WHY MACHIAVELLI WORKS

CHAPTER 1 — ORGANIZATIONAL SURVIVAL AND POLITICAL INSIGHT

The investigation began with a familiar fact: Machiavelli remains intelligible far beyond the political circumstances for which *The Prince* was written.

One explanation is that political behavior has changed less than political institutions. Another is that human dispositions concerning ambition, fear, loyalty, deception, and power remain sufficiently stable for observations about them to retain their force.

The preceding evaluations support another explanation.

Machiavelli repeatedly observes political events through relations that can remain evaluable after the political particulars through which he encounters them have been removed.

The important unit of survival is therefore not necessarily the maxim.

"Whoever causes another to become powerful brings about his own ruin" does not survive literally as a general organizational law. Louis XII supplies a political case involving the intentional strengthening of another political actor.

AC Weakening removes Louis, Alexander, Cesare, political power, intentional strengthening, and eventually even the requirement that the modification be directed toward the continuation later affected.

What survives is weaker:

An intervention selected for one continuation can produce consequences for another continuation when it alters conditions accessible to both.

The garden can instantiate that relation without resembling Renaissance Italy.

Different Machiavellian analyses yield different weakened relations.

His distinction between one's own and others' arms weakens into a problem of retained availability of required capability.

His discussion of new orders weakens into dependence created or preserved through prior instantiation.

His treatment of *fortuna* and modes of proceeding weakens into condition-relative consequence.

His analysis of stronger and weaker political powers weakens into propagation across relational dependencies.

The particulars disappear while some of the relations remain.

The later evaluations strengthen this explanation by progressively removing intentional agency. Where a relation remains evaluable in evolution or ecology after human intention and common objective have disappeared, its survival cannot be explained solely by persistence of the human motives Machiavelli observed.

This suggests a bounded answer to the title question:

Machiavelli works in part because some of his political insights describe organizational relations expressed through political particulars but not wholly dependent upon those particulars.

That claim is importantly weaker than saying that Machiavelli discovered universal laws of organization.

The evaluations establish bounded survival, not universality.

But survival helps explain the unusual durability of his political analysis. A reader can remove prince, principality, mercenary army, Renaissance Italy, political intention, and eventually intentional agency altogether without necessarily removing every dependency relation Machiavelli exposed.

His political insight is therefore partly organizational insight.

CHAPTER 2 — THE MORAL REPUTATION OF MACHIAVELLI

This result permits Machiavelli's familiar moral reputation to be reconsidered without either condemning or rehabilitating him in advance.

The conventional problem is easy to state. Machiavelli appears willing to recommend deception, coercion, cruelty, broken promises, manipulation, and violence when they contribute to political preservation.

That appearance is not an illusion produced by the present analysis. *The Prince* contains genuinely disturbing prescriptions.

But AC exposes a prior question.

When Machiavelli presents an action as necessary, necessary relative to what continuation?

Let:

C_p = continuation of the prince's possession or control of the state;

C_o = a larger organizational continuation affected by that control.

Then:

$N(X, C_p, K)$ does not entail $N(X, C_o, K)$.

Nor does:

$N(X, C_p, K)$ entail $A(X, C_o, K)$.

A measure can be effective, or even necessary, relative to the continuation Machiavelli is analyzing while remaining inadmissible relative to another continuation that a reader regards as morally or politically controlling.

Even within C_p, the relations must remain distinct:

$E(X, C_p, K)$ does not entail $N(X, C_p, K)$.

An action's effectiveness for preserving the prince does not establish that no admissible alternative could preserve the same continuation.

Some apparent disagreements with Machiavelli are therefore structurally ambiguous.

The reader may hear:

X is necessary.

Machiavelli may have established something closer to:

$N(X, C_p, K)$.

The reader then evaluates X relative to C_o, liberty, justice, institutional legitimacy, individual welfare, or another continuation and concludes that X is inadmissible.

Those conclusions need not contradict one another.

X can be necessary relative to C_p and inadmissible relative to C_o.

The error occurs only when necessity established for C_p is promoted without argument into necessity or admissibility for C_o.

This does not establish that Machiavelli is moral. Nor does it establish that he is amoral.

It establishes that the moral question cannot be resolved merely by determining whether an action contributes to the political continuation Machiavelli examines.

The analysis therefore replaces:

Was Machiavelli right to recommend X?

with two questions:

Relative to what continuation and conditions is X effective or necessary in Machiavelli's analysis?

and:

Is X admissible relative to the continuation and conditions under which we wish to evaluate it?

The first is primarily analytical.

The second may be moral, political, institutional, or organizational.

AC does not supply the morally controlling continuation merely by separating these questions. It requires that continuation to be specified and defended rather than silently substituted for Machiavelli's.

Confusing the questions makes Machiavelli easier either to condemn or to excuse than to understand.

The analysis does not identify one continuation that must govern every evaluation. That is not an omitted final step that can be supplied by organizational description alone. Selection of a controlling continuation is a separate evaluative problem requiring specification of whose or which continuation is to govern, what other continuations remain relevant, and what conditions and alternatives are available.

CHAPTER 3 — LIMITS

The argument must stop short of several stronger conclusions.

Cross-Domain Survival Does Not Establish Universality

A proposition surviving evaluation in political, medical, software, engineering, technological, artistic, evolutionary, and ecological settings has survived heterogeneous tests.

It has not thereby been shown to hold in every organization or under every condition.

Downstream Results Do Not Belong Automatically to Machiavelli

The method proceeds in one direction:

Machiavellian text → relational derivation → AC Weakening → frozen proposition → cross-domain evaluation → result.

A result first exposed by ecology, software, cancer treatment, measurement standards, or another cross-domain case belongs initially to that case.

It may be attributed backward to Machiavelli only if it can be independently derived from his text.

Thus the recursive adoption mechanism exposed through VHS/Betamax should not be credited to Machiavelli merely because a weaker relational proposition derived from Chapter III led into it.

Likewise, ecological reconstruction of later conditions and organization without common objective are not thereby Machiavellian propositions.

Weakening Does Not Guarantee Survival

AC Weakening does not establish that every Machiavellian observation contains a domain-independent organizational relation.

Some observations may depend essentially upon political authority, intentional agency, human psychology, historical circumstances, or other particulars.

Removing those particulars may destroy the relation.

That is an acceptable stopping condition.

Non-Instantiation Is a Valid Result

The Hyperrealism/Expressionism evaluation demonstrates this directly.

Difference between the two pictorial organizations did not instantiate the frozen threat relation under K. Threat and competition became available only after an additional allocation relation produced K′.

The proposition was not rewritten to make the initial case fit.

Therefore:

Survival is not guaranteed by the method, and evaluability does not require positive instantiation.

Organizational Description Does Not Establish Admissibility

Discovering that an intervention preserves C does not establish that C ought to be preserved, that X is the only means of preserving it, or that damage imposed upon other continuations is justified.

Thus:

E(X, C, K) does not entail N(X, C, K).

And:

N(X, C, K) does not entail A(X, C, K).

Even:

$A(X, C_1, K)$

does not automatically establish:

$A(X, C_2, K)$.

Cross-Domain Evaluation Is Not Independent Confirmation

The Part III cases were selected and reconstructed for cross-domain evaluation. They were not blind tests, preregistered experiments, or independently generated confirmations.

Their heterogeneity makes successful survival informative, but "independent confirmation" would overstate what the method produced.

The defensible description is:

disciplined cross-domain evaluation under frozen propositions.

Generality of the Relational Forms Remains Provisional

The relational forms identified in Part IV remain candidates for further evaluation rather than demonstrated universal laws.

Further testing may preserve them, restrict them, divide them, or show that their apparent survival depended upon an overly permissive reconstruction.

The strongest conclusion supported here is consequently bounded:

Some relations derived from Machiavelli's political analyses survive disciplined removal of their political particulars and remain evaluable in structurally heterogeneous domains.

That is enough.

The argument does not require more.

CHAPTER 4 — RETURN TO THE FLOCK

Return finally to the sheep.

The wolves are still present.

The sheep still require protection.

The fence still constrains them.

The dogs still patrol.

And the shepherd still appears, at first glance, to occupy Machiavelli's position.

But the story now looks different.

At the beginning, the governing question seemed straightforward:

Does the intervention work?

Electrified night penning stopped nighttime depredation in the reported Oregon case. That result made the intervention appear to supply the answer.

It did not.

We now know to ask which continuation was preserved.

Let:

C_n = prevention of nighttime depredation;

C_F = continuation of the flock;

C_G = continuation of the governing arrangement.

The reported result directly supports effectiveness relative to C_n under the observed conditions:

$E(X, C_n, K_1)$.

It does not by itself establish:

$E(X, C_F, K_1)$,

much less:

$N(X, C_F, K_1)$.

Prevention of nighttime attacks is relevant to continuation of the flock, but the relation between the two must be established rather than silently identified.

Likewise, even if a measure were necessary for C_G, it would not thereby be necessary for C_F:

$N(X, C_G, K)$ does not entail $N(X, C_F, K)$.

We also know to ask whether the conditions remain the same.

The wolves changed their behavior, and daytime depredation increased. The condition set confronting the flock therefore changed:

$K_1 \rightarrow K_2$.

Effectiveness under the earlier conditions does not automatically transfer:

$E(X, C_F, K_1)$ does not entail $E(X, C_F, K_2)$.

The pen may remain effective at preventing nighttime attacks while becoming insufficient for the broader continuation of the flock under the altered behavior of the wolves.

We know not to infer unlimited constraint from successful constraint.

It protects

does not entail:

More of it protects more.

The relation between intervention magnitude and preservation need not be monotonic.

The pen cannot simply become smaller and smaller. At some point increased constraint can damage dependencies belonging to the continuation it was introduced to preserve.

And we know something else now.

The shepherd, wolves, sheep, dogs, fence, and Oregon landscape were never themselves the proposition.

They were one organization through which some of the relations could be seen.

The investigation subsequently removed political authority, coercion, intentional competition, human decision, and eventually common objective.

Some relations disappeared.

Some required more careful specification.

Some did not instantiate.

Some appeared only after additional conditions were introduced.

Others survived.

The flock therefore looks less like an analogy for Machiavelli than it did at the beginning.

That is precisely the point.

The investigation began by asking where Machiavelli was in the story.

The answer appeared obvious:

Machiavelli was the shepherd.

After the weakening and testing, that answer is no longer sufficient.

What survived was not the shepherd.

It was the relation.

The shepherd was one place where a relation could be seen.

Machiavelli’s prince was another.

The garden, the patient, the software interpreter, the measurement standard, the video format, the pictorial practice, the evolving population, and the kelp forest provided settings in which different weakened relations could be evaluated.

They do not all instantiate one relation.

Some do not instantiate the relation for which they were examined.

None of them is Machiavellian merely because a relation derived from Machiavelli can be evaluated there.

And Machiavelli did not anticipate them.

He did something narrower and, perhaps, more remarkable.

He observed political organizations closely enough that some of the relations he exposed remained intelligible and evaluable after politics itself had been removed.

That is one reason Machiavelli still works.